

PixelFlow-OpenAPI

Version	Description	Release Date	Note
V1.0.0	First release	2025-04-30	
V2.0.0	- Added the screen-specific logic and removed the selected screen logic. - Added the complete and relative presets. - Added the connector module protocol. - Added the layer module protocol. - Added the gallery module protocol. - Added the screen chapter 4.1 and preset chapter 6.1, adapted for device versions earlier than V1.7.0. - Added the screen chapter 4.2 and preset chapter 6.2, adapted for devices versions V1.7.0 and later.	2025-07-28	Supported devices (P10, P20, Q8)
V2.0.1	Added the supported device model P80.	2025-10-14	Supported devices (P10, P20, Q8, P80)

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Instructions:

Protocol communication format: <http://ip:8088/unico/url>

IP: The IP address of the controlled device

URL: The corresponding function path

1. Authentication

1.1 User Login

● Basic Information

Path: /v1/system/auth/login

Method: POST

Description:

User login requires a user name and password.

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
username	string	Yes		User name for login	mock: admin
password	string	Yes		Password for login	mock: MTIzNDU2

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	number	Yes			mock: 0
message	string	Yes			

Name	Type	Required	Default Value	Remarks	More
data	object	Yes			
└─ token	string	Yes			

1.2 Read Node Public Information

● Basic Information

Path: /v1/node/open-detail

Method: GET

Description:

● Request Parameters

Query

Name	Required	Example	Remarks
nodeId	No		Node ID

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes	200	Error code	mock: 200
message	string	Yes		ok Error description	mock: ok
data	object	Yes		Business data	
└─ sn	string	Yes		Serial number (SN)	mock: ewrqwerqwe
└─ startTime	string	Yes		Node startup time (timestamp, in milliseconds)	

2. Device

2.1 Read Node Information

● **Basic Information**

Path: /v1/node/detail

Method: GET

Description:

Working mode of the node (indicated by bit position)

7	6	5	4	3	2	1	0	Description	Device Example
	ByPass	Decoder	Player	Video wall splicer	Switcher	Controller	Fiber converter		
							√	Fiber converter	
	√					√	√	MX series	
					√			QD	
						√		Taurus series multimedia player	TU/TB
				√	√			NP	P10/20
				√		√		H series	H

7	6	5	4	3	2	1	0	Descript ion	Device Examp le
	√				√	√	√	All-in- One	K/V/V C/VX
		√						Distribut ed encoder and decoder	

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Authorization		Yes		token

Query

Name	Required	Example	Remarks
nodeId	Yes		Node ID

● Return Data

Name	Type	Requir ed	Defaul t Value	Remarks	More
code	integer	Yes	200	Error code	mock: 200
message	string	Yes		ok Error description	mock: ok
data	object	Yes		Business data	
└─ nodeId	integer	Yes		Node ID	mock: 0
└─ configInfo	object	Yes		Device capability configuration file information	
└─ md5	string	Yes		md5 value of the	

Name	Type	Required	Default Value	Remarks	More
				device capability configuration file	
└─ modeld	integer	Yes		The node's modeld, namely the modeld of the control card on the node	mock: 29974
└─ status	integer	Yes	1	Node operating status: -1: Device illegal. 0: Loading. 1: Loaded, system idle. 2: Loaded, updating. 3: Loaded, creating project file. 4: Loaded, applying project file. 5: Loaded, switching device mode. 6: Loaded, in factory reset. 7: Loaded, saving image. 8: Loaded, compressing logs. 9: Loaded, synchronizing	Max: 2 Min: 0 Enumeration: 0, 1, 2 mock: 1

Name	Type	Required	Default Value	Remarks	More
				backup data. 10: Loaded, in device diagnosis. 11: HDR activated. 12: Loaded, saving cabinets. 13: Loaded, reloading cabinet parameters. 14: Loaded, capturing image. 15: Loaded, in Take process. Only applicable when in Take, Cut and T-Bar at 100%. 16: Loaded, updating the receiving cards. 17: Loaded, refreshing the receiving cards. 18: Loaded, the receiving card sending configuration file. 19: Loaded, converting internal source images. 20: Loaded, in cabinet rotation.	

Name	Type	Required	Default Value	Remarks	More
				21: Loaded, displaying test pattern for correction. 22: Loaded, saving cabinet parameters to hardware. 23: Loaded, applying preset. 24: Loaded, uploading cabinet image quality file. 25: Loaded, in online calibration. 26: Loaded, managing calibration coefficients. 27: Loaded, connecting cabinet tool. 28: Loaded, importing NCP file. 29: Loaded, deleting NCP file. 30: Loaded, exporting NCP file. 31: Loaded, format painter in	

Name	Type	Required	Default Value	Remarks	More
				use. 32: Loaded, rebooting the receiving card. 33: Loaded, device busy, substatus checking needed. 34: Loaded, switching layer specifications. 35: Device in standby mode. 38: Loaded, setting cabinet size. 39: Loaded, configuring connector business with long time consumption. 40: Loaded, switching loading mode. 41: Loaded, verifying primary and backup links. 42: Loaded, in dehumidification mode.	
└─ subStatus	integer []	Yes		Definition of node substatus (multiple	Item type: integer

Name	Type	Required	Default Value	Remarks	More
				statuses may exist simultaneously): 1: Executing one-control-multiple by device. 2: Executing KeyFrame on preset. 3: Previewing KeyFrame execution. 4: Executing video link test. 5: Executing front panel button test. 6: Executing device diagnosis.	
└─		No		Substatus	
└─ sn	string	Yes		Serial number (SN)	mock: ewrqrwerqwe
└─ version	string	Yes	v12.1	Version number	mock: v1.0.0
└─ general	object	Yes		General information	
└─ name	string	Yes	system1	Node name	mock: system1
└─ versionMatching	object	Yes		Version matching information	
└─ serverVersion	string	Yes		userver program	

Name	Type	Required	Default Value	Remarks	More
				version	
└─ minMatchingVersion	string	Yes		Minimum supported frontend version	
└─ minProductVersion	string	No		Minimum supported product version	
└─ powerState	number	Yes		Device status when AC back: 1: Auto power on. 2: Remain off. 3: Restore previous status.	
└─ area	object	Yes		Region information, geographic location	
└─ areaId	integer	Yes	0	Region ID	mock: 1
└─ areaName	string	Yes	name1	Region name	mock: areaName1
└─ mode	object	Yes		Node mode	
└─ workMode	integer	Yes		Node working mode. Refer to the working mode description.	
└─ masterSlave	integer	Yes		Master and slave node: 0: Independent	

Name	Type	Required	Default Value	Remarks	More
				node. 1: Master node. 2: Slave node.	
└─ location	object	Yes		Node location	
└─ enable	number	Yes		Switch: 0: Disable. 1: Enable.	
└─ time	object	Yes		Grant time	
└─ timestamp	integer	Yes		Timestamp (the total number of seconds from January 1, 1970 (00:00:00 GMT) to the current time)	
└─ timezone	integer	Yes		Timezone identifier, range 0~89. See the node configuration file definition.	
└─ stateInfo	object	Yes		Monitoring status	
└─ slot	object []	Yes			Min: 3 Elements all unique: true Max: 3 Item type: object

Name	Type	Required	Default Value	Remarks	More
└─ slotId	number	Yes		Slot No.	
└─ status	object []	Yes			Item type: object
└─ type	number	Yes		Monitoring type: When type=0, it indicates power voltage. When type=1, it indicates temperature. When type=2, it indicates fan speed. When type=3, it indicates fan duty cycle. When type=4, it indicates power signal status.	
└─ index	number	Yes		Number: When Type=0, index indicates power number. When Type=1, index indicates sensor number. When Type=2, index indicates fan number. When Type=3, index indicates fan number.	

Name	Type	Required	Default Value	Remarks	More
				When Type=4, index indicates power number.	
└─ state	number	Yes		Status: 0: Abnormal. 1: Normal.	
└─ value	number	Yes		When type=0, value indicates output power voltage. When type=1, value indicates card temperature. When type=2, value indicates fan speed. When type=3, value indicates fan duty cycle. When type=4, value indicates power signal status. [0: Power not connected; 1: Power connected]	
└─ syncTime	object	Yes		Time synchronization	
└─ timeMode	integer	Yes	2	Time grant mode: 0: Disable	Max: 3 Min: 0

Name	Type	Required	Default Value	Remarks	More
				automatic time synchronization. 1: NTP. 2: GPS. 3: LORA.	
└─ gps	object	Yes		GPS time synchronization structure	
└─ lora	object	Yes			
└─ groupId	string	Yes	121212	Group ID. Assigned by users for device grouping (string format, up to 10 bytes, limited by the control computer).	
└─ ntp	object	Yes		NTP time synchronization structure	
└─ server	string	Yes		NTP time synchronization server	
└─ networkType	number	Yes	1	Network type of the current terminal: 0: No network. 1: Wired network. 2: Wi-Fi. 3: 2G. 4: 3G. 5: 4G.	Max: 5 Min: 0 mock: 1

Name	Type	Required	Default Value	Remarks	More
└─ subcards	object []	Yes		List of cards on the node (all card slots)	Min: 1 Elements all unique: true Max: 1 Item type: object
└─ slotId	number	Yes	1	The card's slot No.	mock: 1
└─ modelId	number	Yes	1	The card's modelId	mock: 30977
└─ boardId	number	Yes	1	The card's hardware version	mock: 1
└─ online	number	Yes	1	The card status: 0: The card is offline, meaning the card is removed. 1: The card is online (It is the normal status and default value. Only cards in this status can be updated). 2: The card is inserted, but its link with the control card is abnormal. 3: The card type is not supported by	Max: 1 Min: 0 Enumeration: 1 mock: 1

Name	Type	Required	Default Value	Remarks	More
				this device (For example, an F-series card is inserted into a D-series device). 4: Different types of cards are inserted. For example, an input card is inserted into an output slot. 5. The card is online, but its version does not match the device's program version. 】	
└─ rotate	number	Yes		Whether the card supports connector rotation: 0: Not supported. 1: Supported.	
└─ genlock	object	Yes			
└─ type	number	Yes		Genlock type: 0: Internal sync/Disabled. 1: Sync with the Genlock source. 2: Sync with the input source from the connector.	

Name	Type	Required	Default Value	Remarks	More
└─ sourceId	number	Yes		Reference source ID: When type=1, sourceId=nodeId. When type=2, sourceId=interfaceId.	
└─ phase	object	Yes		Phase offset	
└─ enable	integer	Yes		Sync phase adjustment enable: 0: Disable/off. 1: Enable/on.	
└─ offset	integer	Yes		Sync shift setting. Unit: clk. Range: -40000~+40000.	
└─ internalFrame	object	No		Internal frame rate	
└─ framerate	number	No		Frame rate	
└─ output	object	No		Genlock out	
└─ type	number	Yes		Output sync type: 1: Genlock in 3: Internal frame rate	
└─ genlockStatus	object	Yes		Current Genlock status	
└─ type	number	Yes		Current Genlock status: 0: Not enabled. 1:	

Name	Type	Required	Default Value	Remarks	More
				Adjusting. 2: Sync successful. 3: Sync failed. Frame rate mismatch. 4: Sync failed. Sync signal lost. 5: Sync failed. No available screens. 6: Sync failed. Bandwidth exceeded. 7: Sync failed. Sync frame > 240Hz	
└─ deviceBackup	object	Yes		Device backup	
└─ master	object	Yes		Primary device information	
└─ sn	string	Yes		MAC address	
└─ mac	string	Yes		Device SN	
└─ ip	string	Yes		Device IP	
└─ name	string	Yes		Device name	
└─ modelId	integer	Yes		Node modelId	
└─ slave	object []	Yes		Backup device SN	Item type: object
└─ sn	string	Yes		MAC address	
└─ mac	string	Yes		Device SN	
└─ ip	string	Yes		Device IP	

Name	Type	Required	Default Value	Remarks	More
└─ name	string	Yes		Device name	
└─ modelId	integer	Yes		Node modelId	
└─ deviceBeacon	integer	Yes		Device locate: 0: Disable. 1: Enable.	
└─ deviceType	integer	Yes		Device type: 0: Invalid parameter. 1: Single-card device. 2: Card-based device.	
└─ presetPlayEffect	integer	Yes		Preset transition effect: 0: CUT. 1: FADE.	
└─ virtualId	integer	Yes		The device's virtual ID, namely the device ID in a cascade	
└─ layerMode	number	Yes		Layer specifications mode: 0: Default mode. (Switcher: 0: 2*MAIN+10*PIP; 1: 4*MAIN+4*PIP.)	
└─ authorInfo	object	Yes		Authorization-related information	

Name	Type	Required	Default Value	Remarks	More
└─ startTimeStamp	number	Yes		Authorization start time, timestamp, in milliseconds	
└─ durationTimeStamp	number	Yes		Authorization duration, timestamp, in milliseconds	
└─ flag	integer	Yes		0: Permanent authorization. 1: Temporary authorization. 2: Authorization expired.	
└─ fanMode	number	Yes		0: Unknown. 1: Standard mode. 2: High-performance mode.	
└─ plugin	object	Yes		Plugin information	
└─ resource	object	Yes		Resource evaluation plugin	
└─ url	string	Yes		Plugin download URL	
└─ md5	string	Yes		Plugin md5 value	
└─ version	string	Yes		Plugin version 1.0.0	
└─ mvr	object	Yes		MVR information	
└─ mode	numb	Yes		MVR mode	

Name	Type	Required	Default Value	Remarks	More
	er			0: Full screen 1: Keep ratio	
└─ displaySwitchInfo	object	Yes		Display on/off information.	
└─ startupPictureEnable	integer	Yes		Startup image switch: 0: Disable. 1: Enable. (Default)	
└─ manufacturerIdentityEnable	integer	Yes		Manufacture logo switch: 0: Disable. 1: Enable. (Default)	
└─ antiStatic	object	Yes		Antistatic information	
└─ time	number	Yes		Antistatic duration (unit: ms)	
└─ enable	number	Yes		Antistatic switch: 0: Disable. 1: Enable.	
└─ senderCapacityInfo	object	Yes		Device loading mode Information	
└─ capacity	integer	Yes		Loading mode: 0: Default loading capacity 2: 2.2 million pixels	

2.2 Factory Rest

● Basic Information

Path: /v1/system/restore-factory

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		
Authorization		Yes		token

Body

Name	Type	Required	Default Value	Remarks	More
restoreFactory	object []	Yes		Factory reset	Item type: object
type	number	Yes		The meaning of the para field for different type values: When type=0, para means condition configuration restart. When type=1, para means IP setting configuration. When type=2, para means EDID settings. When type=3, para means gallery data. When type=4, para means language	

Name	Type	Required	Default Value	Remarks	More
				settings. When type=5, para means project file. When type=6, para means preset. When type=7, para means layer preset. When type=8, para means user file. When type=9, para means device mode.	
└─ para	number	Yes		The meaning of para values for different type values: When type=0, para value: 0: Restart device. When type=1, para value: 0: Retain IP settings. When type=2, para value: 0: Retain EDID settings. When type=3, para value: 0: Retain gallery. When type=4, para value: 0: Retain language settings. When type=5, para value: 0: Retain project files. When type=6, para value: 0: Retain presets. When type=7, para value: 0: Retain layer presets. When type=8, para value: 0: Retain user file. When type=9, para value: 0:	

Name	Type	Required	Default Value	Remarks	More
				Retain device mode.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	mock: 200
message	string	Yes		Response message	mock: ok
data	object	Yes			

2.3 Set Transition Effect

● Basic Information

Path: /v1/screen/global/switch-effect

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
switchEffect	object	Yes		The global transition effect	
└─ time	number	Yes		The transition duration (unit: ms)	
└─ type	number	Yes		Transition method: 0: CUT. 1: FADE.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object []	Yes			Item type: object

2.4 Set Swap

● Basic Information

Path: /v1/screen/global/swap

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		
Authorization		Yes		token

Body

Name	Type	Required	Default Value	Remarks	More
enable	number	Yes		Swap switch: 0: Disable. 1: Enable.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			

2.5 Read Node Monitoring Status

- **Basic Information**

Path: /v1/node/state-info

Method: GET

Description:

- **Request Parameters**

Headers

Name	Value	Required	Example	Remarks
Authorization		Yes		token

Query

Name	Required	Example	Remarks
nodeId	Yes		Node ID

- **Return Data**

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Error code	mock: @integer
message	string	Yes		Error description	mock: @string
data	object	Yes		Business data	
└─ nodeId	integer	Yes		Node ID	
└─ stateInfo	object	Yes		Monitoring status	
└─ slot	object []	Yes			Item type: object
└─	number	Yes		Slot No.	

Name	Type	Required	Default Value	Remarks	More
slotId					
└─ status	object []	Yes			Item type: object
└─ type	number	Yes		Monitoring type: When type=0, it indicates power voltage. When type=1, it indicates temperature. When type=2, it indicates fan speed. When type=3, it indicates fan duty cycle. When type=4, it indicates power signal status.	
└─ index	number	Yes		Number: When Type=0, index indicates power number. When Type=1, index indicates sensor number. When Type=2, index indicates fan number. When Type=3, index indicates fan number. When Type=4, index indicates power number.	
└─ state	number	Yes		Status: 0: Abnormal. 1: Normal.	
└─ value	number	Yes		When type=0, value indicates output power voltage. Calculated by multiplying the	

Name	Type	Required	Default Value	Remarks	More
				actual output voltage by 10 to avoid decimals. When type=1, value indicates card temperature. When type=2, value indicates fan speed. When type=3, value indicates fan duty cycle. When type=4, value indicates power signal status. [0: Power not connected; 1: Power connected]	
 online	number	Yes		The card status: 0: The card is offline, meaning the card is removed. 1: The card is online (It is the normal status and default value. Only cards in this status can be updated). 2: The card is inserted, but its link with the control card is abnormal. 3: The card type is not supported by this device (For example, an F-series card is inserted into a D-series device). 4: Different types of cards are inserted. For example, an input card is inserted into an output slot. 5. The card is online, but its version does not match the device's program version.	

3. Connector

3.1 Read Connector Details

- **Basic Information**

Path: /v1/interface/list-detail

Method: GET

Description:

1. When splitId != 0, this interfaceId signifies the unique ID of the split source of the main source.
2. Optional parameters for the query: if no parameters are provided, all data is returned. If multiple parameters are provided, they must satisfy an & condition.

- **Request Parameters**

Query

Name	Required	Example	Remarks
limit	No		Items per page, ignored if this parameter is absent
page	No		Query page number. Ignored if this parameter is absent.
interfaceId	No		Connector ID (main source and sub-source), unique across all interfaceType, ignored if this parameter is absent
mosicId	No		Mosaic source ID, ignored if this parameter is absent
groupId	No		Group ID, ignored if this parameter is absent
presetId	No		Preset ID, used together with sceneType. Without this parameter, no query is performed. With this parameter, the connector data for the specified preset is queried.
groupType	No		Group type, ignored if this parameter is absent.

Name	Required	Example	Remarks
musicType	No		Mosaic source type. 0: common input. Ignored if this parameter is absent.
sceneType	No		Preset type. Without this parameter, connector information within the preset will not be queried.
interfaceType	No		Connector type, ignored if this parameter is absent
online	No		0: The card is offline, meaning the card is removed. 1: The card is online (It is the normal status and default value. Only cards in this status can be updated). 2: The card is inserted, but its link with the control card is abnormal. 3: The card type is not supported by this device (For example, an F-series card is inserted into a D-series device). 4: Different types of cards are inserted. For example, an input card is inserted into an output slot. Ignored if this parameter is absent.

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	mock: 0
message	string	Yes		Response message	mock: ok
data	object	Yes			
└─ list	object []	Yes			Min: 2 Elements all unique: true Max: 2

Name	Type	Required	Default Value	Remarks	More
					Item type: object
└─ interfaced	number	Yes		Connector ID	Max: 111 Min: 1
└─ state	number	Yes		0: Not connected. 1: Connected, normal. 2: Connected, overloaded. 3: Connected, connector unstable. 4: Connected, not standard. 5: Not connected, not standard. 6: Not connected, overloaded.	
└─ interfacedObj	object	Yes		Presentation of the connector ID in a split format	
└─ nodeId	number	Yes		Node ID	mock: 0
└─ slotId	number	Yes		Slot No.	mock: 0
└─ connectorId	number	Yes		Connector number of the connector	mock: 1
└─ splitId	number	Yes		0 indicates the main source, while other numbers represent the sub-source No. of the connector.	mock: 1

Name	Type	Required	Default Value	Remarks	More
└─ auxiliaryInfo	object	Yes		Auxiliary information	
└─ online	number	Yes		Status of the card where the connector is located: 0: The card is offline, meaning the card is removed. 1: The card is online and the connector connection is normal. 2: The card is online and the connector connection is abnormal.	Enumeration: 0, 1
└─ connectorInfo	object	Yes		Connector property information	
└─ interfaceType	number	Yes		Connector type: 2: Input connector. 4: Output connector. 8: AUX connector. 16: MVR connector. 32: Input view connector. 64: Main output connector.	Enumeration: 0, 1, 2, 3
└─ type	integer	Yes		Connector type enumeration	
└─ workMode	number	Yes		Working mode: 0: Main mode. 1: Copy mode. 2: Disable mode. Distinction in usage between input connectors and output connectors: Input card connectors:	Enumeration: 0, 1, 2, 3

Name	Type	Required	Default Value	Remarks	More
				Main mode, disable mode. Output card connectors: Main mode, copy mode, disable mode.	
└─ copyInfo	object	Yes		When workMode is not in copy mode, the copyInfo field is completely invalid. When workMode is in copy mode, copyInfo indicates which connector is being copied.	
└─ type	integer	Yes		Target type: 0: Input or output connector	Enumeration: 0
└─ id	integer	Yes		Target ID: When type=0, it indicates interfacedId.	mock: id1
└─ capacity	number	Yes		Connector capacity: 0: SL. 1: DL. 2: 4K. 255: Invalid value.	Enumeration: 0, 1, 2, 255
└─ screenInfo	object	Yes		Information about the screen loaded by the connector	
└─ name	string	Yes		Screen name	
└─ group	object	Yes		Connector grouping information	
└─ id	integer	Yes		Group ID	

Name	Type	Required	Default Value	Remarks	More
└─ name	string	Yes		Group name	
└─ physicalGroupID	number	Yes		Physical connector group ID	
└─ alias	string	Yes		Connector alias	
└─ general	object	Yes		Connector's general information	
└─ name	string	Yes		Connector name	mock: generalName1
└─ presetThumbnailTiming	object	Yes		Preset thumbnail timing	
└─ interlaced	number	Yes		Interlaced scanning: 0: Interlaced. 1: Progressive.	
└─ frame	number	Yes		Frame rate (floating-point format)	mock: 1
└─ width	number	Yes		Width	mock: 1920
└─ height	number	Yes		Height	mock: 960
└─ standardType	number	Yes		Resolution standard: 0: No standard. 1: DMT 2: CVT 3: CVT-RBV1 4: CVT-RBV2	

Name	Type	Required	Default Value	Remarks	More
				5: CVT-RBV3 6: CEA 7: SMPTE	
└─ actualThumbTiming	object	Yes		Actual thumbnail timing	
└─ interlaced	number	Yes		Interlaced scanning: 0: Interlaced. 1: Progressive.	
└─ frame	number	Yes		Frame rate (floating-point format)	mock: 2
└─ width	number	Yes		Width	mock: 1920
└─ height	number	Yes		Height	mock: 960
└─ standardType	number	Yes		Resolution standard: 0: No standard. 1: DMT 2: CVT 3: CVT-RBV1 4: CVT-RBV2 5: CVT-RBV3 6: CEA 7: SMPTE	
└─ presetTiming	object	Yes		Connector preset timing details	
└─ interlaced	number	Yes		Interlaced scanning: 0: Interlaced. 1: Progressive.	Enumeration: 0, 1

Name	Type	Required	Default Value	Remarks	More
└─ refresh	number	Yes		Frame rate (floating-point format)	
└─ horizontal	object	Yes		Horizontal timing	
└─ addTime	number	Yes		Visible width (H active, namely the commonly referred resolution width)	mock: 1920
└─ frontPorch	number	Yes		Front porch	mock: 1
└─ sync	number	Yes		Sync value	mock: 1
└─ syncPolarity	number	Yes		Sync polarity: 0: Negative. 1: Positive.	Enumeration: 0, 1
└─ totalTime	number	Yes		Total width (H total)	mock: 1920
└─ vertical	object	Yes		Vertical timing	
└─ addTime	number	Yes		Visible height (V active, namely the commonly referred resolution height)	mock: 960
└─ frontPorch	number	Yes		Front porch	mock: 1
└─ sync	number	Yes		Sync value	mock: 2
└─ syncPolarity	number	Yes		Sync polarity: 0: Negative. 1: Positive.	Enumeration: 0, 1

Name	Type	Required	Default Value	Remarks	More
└─ totalTime	number	Yes		Total height (V total)	mock: 960
└─ standardType	number	Yes		Resolution standard: 0: No standard. 1: DMT 2: CVT 3: CVT-RBV1 4: CVT-RBV2 5: CVT-RBV3 6: CEA 7: SMPTE	
└─ actualTiming	object	Yes		Connector actual timing details	
└─ interlaced	number	Yes		Interlaced scanning: 0: Interlaced. 1: Progressive.	Enumeration: 0, 1
└─ refresh	number	Yes		Frame rate (floating-point format)	mock: 60
└─ horizontal	object	Yes		Horizontal timing	
└─ addTime	number	Yes		Visible width (H active, namely the commonly referred resolution width)	mock: 1960
└─ frontPorch	number	Yes		Front porch	mock: 1
└─ sync	number	Yes		Sync value	mock: 1
└─ syncPolarity	number	Yes		Sync polarity: 0: Negative.	Enumeration: 0, 1

Name	Type	Required	Default Value	Remarks	More
				1: Positive.	
└─ totalTime	number	Yes		Total width (H total)	mock: 1960
└─ vertical	object	Yes		Vertical timing	
└─ addTime	number	Yes		Visible height (V active, namely the commonly referred resolution height)	mock: 960
└─ frontPorch	number	Yes		Front porch	
└─ sync	number	Yes		Sync value	
└─ syncPolarity	number	Yes		Sync polarity: 0: Negative. 1: Positive.	Enumeration: 0, 1
└─ totalTime	number	Yes		Total height (V total)	mock: 960
└─ standardType	string	Yes		Resolution standard: 0: No standard. 1: DMT 2: CVT 3: CVT-RBV1 4: CVT-RBV2 5: CVT-RBV3 6: CEA 7: SMPTE	
└─ timingSelect	object	Yes		Connector's timing type	

Name	Type	Required	Default Value	Remarks	More
└─ select	number	Yes		Timing type: 0: presetThumbTiming. 1: presetTiming. 2: edidFile.	Enumeration: 0, 1, 2
└─ imageQuality	object	Yes			
└─ imageQualityMode	number	Yes		Image quality mode (image quality preset): 0: Standard mode. 1: Document mode. 2: Conference mode. 3: Video mode.	
└─ eyeCare	number	Yes		Eye comfort mode switch: 0: Disable. 1: Enable.	
└─ hue	number	Yes		Hue	
└─ saturation	number	Yes		Saturation	
└─ gamma	number	Yes		Gamma value in image quality. Gamma value [1.00~4.00]*100, to two decimal places.	
└─ gammaScale	number	Yes		Scale ratio multiplier: 1: No scaling. 100: Scaling down by a factor of 100.	
└─ contrast	object	Yes		Contrast	

Name	Type	Required	Default Value	Remarks	More
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ coefficient	number	Yes		Coefficients	
└─ brightness	object	Yes		Brightness	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ coefficient	number	Yes		Coefficients	
└─ shadow	object	Yes		Shadow	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ W	number	Yes		Black level component	

Name	Type	Required	Default Value	Remarks	More
 highLight	object	Yes		Highlight	
 R	number	Yes		Red component	
 G	number	Yes		Green component	
 B	number	Yes		Blue component	
 W	number	Yes		White level component	
 colorTemperature	object	Yes		Color temperature	
 mode	number	Yes		Adjustment mode: 0: Quick mode. 1: Advanced mode.	
 quickAdjust	object	Yes		Quick parameter adjustment	
 value	number	Yes		Color temperature range: 1700~15000, default value: 6500, step: 100, unit: K	
 advanceAdjust	object	Yes		Advanced parameter adjustment	
 R	number	Yes		R component: 0~255, default: 255, step: 1, unit: none	
 G	number	Yes		G component: 0~255, default: 255, step: 1, unit:	

Name	Type	Required	Default Value	Remarks	More
				none	
└─ B	number	Yes		B component: 0~255, default: 255, step: 1, unit: none	
└─ isMonochrome	number	Yes		Whether to enable Monochrome: 0: No. 1: Yes.	
└─ isInverse	number	Yes		Whether to invert colors: 0: No. 1: Yes.	
└─ crop	object	Yes		Connector cropping information. To ensure that the cropping ratio is not affected by the actual size of the source after connector cropping, a cropping ratio is used. The ratio is magnified 100,000 times because the current maximum width is 8192, corresponding to a minimum crop of 64. A magnification greater than $8192/64=128$ times ensures an error of less than 1. A magnification of 100,000 times can accommodate a resolution of 640,000. "widthResolution"	

Name	Type	Required	Default Value	Remarks	More
				represents the resolution width of the current connector, and "heightResolution" represents the resolution height of the current connector. "width" represents the actual cropped width, and "height" represents the actual cropped height. For calculations involving width, height, and xy coordinates, rounding is used.	
 enable	number	Yes		Crop enable: 0: Disable. 1: Enable.	Enumeration: 0, 1
 widthRatio	number	Yes		Cropping width ratio. $\text{widthRatio} = \text{round}(\text{width}/\text{widthResolution} * 100000)$.	mock: 0.3
 heightRatio	number	Yes		Cropping height ratio. $\text{heightRatio} = \text{round}(\text{height}/\text{heightResolution} * 100000)$.	mock: 0.4
 xRatio	number	Yes		Cropping's x position ratio. $\text{xRatio} = \text{round}(\text{x}/\text{widthResolution} * 100000)$.	mock: 0.2

Name	Type	Required	Default Value	Remarks	More
└─ yRatio	number	Yes		Cropping's y position ratio. yRatio = round (y/heightResolution * 100000).	mock: 0.3
└─ hdcp	object	Yes		HDCP settings	
└─ enable	number	Yes		Enable status: 0: Disable. 1: Enable.	Enumeration: 0, 1
└─ preset	number	Yes		Preset type: 0: Off. 1: HDCP 1.4. 2: HDCP 1.3. 3: HDCP 2.2.	Enumeration: 0, 1, 2, 3
└─ actual	number	Yes		Actual type: 0: Off. 1: HDCP 1.4. 2: HDCP 1.3. 3: HDCP 2.2. 4: HDCP not recognized.	
└─ colorSpace	object	Yes		Color space	
└─ actual	number	Yes		Actual color space: 0: RGB 4:4:4. 1: YCbCr 4:2:2. 2: YCbCr 4:4:4. 3: YCbCr 4:2:0.	Enumeration: 0, 1, 2, 3
└─ preset	number	Yes		Preset color space: 0: RGB 4:4:4. 1: YCbCr 4:2:2.	Enumeration: 0, 1, 2, 3

Name	Type	Required	Default Value	Remarks	More
				2: YCbCr 4:4:4. 3: YCbCr 4:2:0. 255: AUTO.	3, 255
 colorDepth	object	Yes		Bit depth	
 actual	number	Yes		Actual bit depth: 8: 8Bit. 10: 10Bit. 12: 12Bit. 16: 16Bit.	Enumeration: 8, 10, 12, 16
 preset	number	Yes		Preset bit depth: 8: 8Bit. 10: 10Bit. 12: 12Bit. 16: 16Bit. 255: AUTO.	Enumeration: 8, 10, 12, 16, 255
 colorRange	object	Yes		Color range	
 actual	number	Yes		Actual color range: 0: LIMIT. 1: FULL.	Enumeration: 0, 1
 preset	number	Yes		Preset color range: 0: LIMIT. 1: FULL. 255: AUTO.	Enumeration: 0, 1, 255
 colorGamut	object	Yes		Color gamut	
 actual	number	Yes		Actual color gamut: 0: Rec.601.	Enumeration: 0, 1, 2,

Name	Type	Required	Default Value	Remarks	More
				1: Rec.709. 2: DCI-P3. 3: Rec.2020. 4: NTSC. 5: aRGB. 254: Unknown.	3, 4, 5
└─ preset	number	Yes		Preset color gamut: 0: Rec.601. 1: Rec.709. 2: DCI-P3. 3: Rec.2020. 4: NTSC. 5: aRGB. 255: AUTO.	Enumeration: 0, 1, 2, 3, 4, 5
└─ colorDynamicRange	object	Yes		Dynamic range	
└─ actual	number	Yes		Actual dynamic range: 0: SDR. 1: HDR10. 2: HLG.	Enumeration: 0, 1, 2
└─ preset	number	Yes		Preset dynamic range: 0: SDR. 1: HDR10. 2: HLG. 255: AUTO.	Enumeration: 0, 1, 2, 255
└─ testPattern	object	Yes		Test pattern	
└─ enable	number	Yes		Enable: 0: Disable.	

Name	Type	Required	Default Value	Remarks	More
				1: Enable.	
└─ type	number	Yes		Test pattern type: 0x0000: Black. 0x0001: Red. 0x0002: Green. 0x0003: Blue. 0x0004: White. 0x0005: Vertical stripes. 0x0006: Horizontal stripes. 0x0007: Checkerboard. 0x0100: Horizontal red gradient. 0x0101: Horizontal green gradient. 0x0102: Horizontal blue gradient. 0x0103: Horizontal white gradient. 0x0104: Vertical red gradient. 0x0105: Vertical green gradient. 0x0106: Vertical blue gradient. 0x0107: Vertical white gradient. 0x0200: Horizontal white lines. 0x0201: Vertical white lines. 0x0202: White diagonal left. 0x0203: White diagonal right.	

Name	Type	Required	Default Value	Remarks	More
				0x0204: Crossed lines. 0x0205: Crossed diagonals. 0x0300: Locate. 0xFFFE: Custom test pattern. 0xFFFF: Off.	
└─ bright	number	Yes		Brightness	
└─ grid	number	Yes		Its meaning is determined by the field "gridType". If this field is absent, it indicates the grid density.	
└─ gridType	number	Yes		Description of grid unit: 0: "grid" indicates grid density, adjustable in 8 levels ranging from 1 to 8, with a default of level 5. 1: "grid" indicates grid spacing in pixels, with a minimum of 1 pixel, a maximum equal to 1/4 of the horizontal maximum pixel count for the configured connector, a step size of 1, and a default of 32.	
└─ gridLineWidth	number	Yes		Grid width, unit: pixel	
└─ speed	number	Yes		Speed	

Name	Type	Required	Default Value	Remarks	More
 smpteMapping	object	Yes		smpteMapping settings of 3G-SDI connector	
 actual	number	Yes		SMPTE 425M Mapping: 0: Invalid. 1: Level A. 2: Level B. Currently, only SDI connectors support Level A/Level B settings; other connectors default to unused.	Enumeration: 0, 1, 2
 preset	integer	Yes		SMPTE 425M Mapping: 0: Unused. 1: Level A. 2: Level B. Currently, only SDI connectors support Level A/Level B settings; other connectors default to unused.	Enumeration: 0, 1, 2
 removeInterlace	object	Yes		Connector automatic deinterlacing	
 enable	integer	Yes		Enable status: 0: Disable. 1: Enable.	Enumeration: 0, 1 mock: @integer
 cutout	object	Yes		Connector DSK	

Name	Type	Required	Default Value	Remarks	More
 enable	integer	Yes		DSC enable: 0: Disable. 1: Enable.	Enumeration: 0, 1
 cursor	object	Yes		Cursor position	
 enable	number	Yes		Show cursor: 0: Disable. 1: Enable.	Enumeration: 0, 1
 pos	object	Yes		Cursor coordinates (using the top-left corner of the visible area as the origin)	
 x	integer	Yes		Pick point x-coordinate	mock: 122
 y	integer	Yes		Pick point y-coordinate	mock: 0
 color	object	Yes		Picked color	
 R	number	Yes		Red component	mock: 255
 G	number	Yes		Green component	mock: 255
 B	number	Yes		Blue component	mock: 0
 type	integer	Yes		DSC type: 0: Luma key. 1: Chroma key. 2: Smart key.	Enumeration: 0, 1
	object	Yes		Luma key	

Name	Type	Required	Default Value	Remarks	More
brightness	t				
└─ threshold	integer	Yes		Clip (brightness threshold), range: 0~100	Max: 100 Min: 0
└─ gain	integer	Yes		Brightness gain, range: 0~100	Max: 100 Min: 0
└─ foregroundColor	object	Yes		Foreground color configuration	
└─ enable	integer	Yes		Foreground color enable: 0: Disable. 1: Enable.	Enumeration: 0, 1
└─ R	integer	Yes		Red component	mock: 255
└─ G	integer	Yes		Green component	mock: 0
└─ B	integer	Yes		Blue component	mock: 0
└─ hue	object	Yes		Chroma key	
└─ color	object	Yes		Key color	
└─ R	integer	Yes		Red component	mock: 255
└─ G	integer	Yes		Green component	mock: 0
└─ B	integer	Yes		Blue component	mock: 0

Name	Type	Required	Default Value	Remarks	More
└─ pos	object	Yes		Cursor coordinates (using the top-left corner of the visible area as the origin)	
└─ x	number	Yes		Pick point x-coordinate	
└─ y	number	Yes		Pick point y-coordinate	
└─ hueClip	integer	Yes		Hue clip, range: 5~hueRamp	Max: 6 Min: 5
└─ hueRamp	integer	Yes		Hue Ramp, range: 5~180	Max: 180 Min: 5
└─ saturationClip	integer	Yes		Saturation clip, range: 5~100	Max: 100 Min: 5
└─ saturationGain	integer	Yes		Gain (softness threshold), range: 5~100	Max: 100 Min: 5
└─ spill	integer	Yes		Spill, range: 5~100	Max: 100 Min: 5
└─ shadow	integer	Yes		Shadow, range: 5~100	Max: 100 Min: 5
└─ highLigh	integer	Yes		Highlight, range: 5~100	Max: 100 Min: 5
└─ edgeSmooth	integer	Yes		Edge smooth: 0: Disable. 1: Enable.	Enumeration: 0, 1

Name	Type	Required	Default Value	Remarks	More
└─ intelligence	object	Yes		Smart key	
└─ color	object	Yes		Picked color	
└─ R	string	Yes		Red component	
└─ G	string	Yes		Green component	
└─ B	string	Yes		Blue component	
└─ pos	object	Yes		Cursor coordinates (using the top-left corner of the visible area as the origin)	
└─ x	integer	Yes		Pick point x-coordinate	
└─ y	integer	Yes		Pick point y-coordinate	
└─ mattingStrength	integer	Yes		Keying strength	
└─ gainAdjust	integer	Yes		Gain adjustment	
└─ hdr	object	Yes		Connector HDR properties	
└─ sdr	object	Yes		SDR conversion settings	
└─ gamma	number	Yes		Gamma value, range: 1.00~4.00	Max: 4 Min: 1
└─ hdr10	object	Yes		HDR10 conversion settings	

Name	Type	Required	Default Value	Remarks	More
└─ peakLuminance	number	Yes		Peak screen brightness, range: 0~10000	Max: 10000 Min: 0
└─ envLuminance	number	Yes		Screen's ambient brightness, range: 0~10000	Max: 10000 Min: 0
└─ pqMode	number	Yes		Select PQ curve: 0: ST2084. 1: ST2086.	Enumeration: 0, 1
└─ toneMapping	number	Yes		Curve adjustment switch: 0: Disable. 1: Enable.	Enumeration: 0, 1
└─ maxccl	object	Yes		Peak source brightness	
└─ enable	number	Yes		Custom MaxCLL enable: 0: Disable. 1: Enable.	
└─ presetMaxccl	number	Yes		Custom peak source brightness, range: 0~10000	Max: 10000 Min: 0
└─ hlg	object	Yes		HLG conversion settings	
└─ peakLuminance	number	Yes		Peak brightness, range: 100~10000	Enumeration: 1, 2, 3, 4, 5, 6, 7
└─ envLuminance	number	Yes		Ambient brightness	Max: 100

Name	Type	Required	Default Value	Remarks	More
					Min: 0
└─ keyPosition	integer []	Yes		The key position index, starting from 0	Item type: integer
└─		No		The position sequence value	
└─ metadata	object []	Yes		metadata data	Item type: object
└─ type	number	Yes		The meaning of the para field for different type values: When type==0, para meaning: Electro-optical conversion (electroOpticalTransferFunction). When type==1, para meaning: White point x-coordinate (whitePointX). When type==2, para meaning: White point y-coordinate (whitePointY). When type==3, para meaning: MaxDML (maxDisplayMasteringLuminance). When type==4, para meaning: MinDML (minDisplayMasteringLuminance). When type==5, para meaning: MaxCLL	

Name	Type	Required	Default Value	Remarks	More
				(maximumContentLightLevel). When type==6, para meaning: MaxFLL (maximumFrameAverageLightLevel).	
└─ para	number	Yes		The meaning of para values for different type values: When type=0, para value: 0: SDR. 1: HDR. 2: SMPTE 2084. 3: HLG. When type==1, para value: White point x-coordinate (whitePointX). When type==2, para value: White point y-coordinate (whitePointY). When type==3, para value: MaxDML (maxDisplayMasteringLuminance). When type==4, para value: MinDML (minDisplayMasteringLuminance). When type==5, para value: MaxCLL (maximumContentLightLevel). When type==6, para value: MaxFLL (maximumFrameAverageLightLevel).	
└─ ledHdr	object	Yes			

Name	Type	Required	Default Value	Remarks	More
	t				
└─ dynamicRange	number	Yes		Dynamic range: 0: SDR. 1: HDR. 2: HLG. 255: From input.	
└─ pqMode	number	Yes		PQ mode: 0: ST2084 (PQ). 1: ST2086 (Linear mapping).	
└─ maxcllOverride	number	Yes		MaxCLL override switch: 0: Disable. 1: Enable.	
└─ maxCll	number	Yes		Max content light level (MaxCLL), range:100~10000	
└─ streamInfo	object	Yes		Optical port's stream pulling information	
└─ destIp	string	Yes		Destination IP	
└─ port	number	Yes		Port	
└─ srcIp	string	Yes		Source IP	
└─ colorSpace	number	Yes		Color space in the optical port's stream pulling information	
└─ colorDepth	number	Yes		Bit depth in the optical port's stream pulling information	

Name	Type	Required	Default Value	Remarks	More
└─ netInfo	object	Yes		Optical port IP	
└─ dhcp	number	Yes		Whether to enable DHCP: 0: Static IP. 1: Dynamic IP.	
└─ ip	string	Yes		Connector IP	
└─ netmask	string	Yes		Subnet mask	
└─ gateway	string	Yes		Gateway	
└─ mac	string	Yes		Ethernet port's MAC address	
└─ fileInfo	object	Yes		File information imported to connectors	
└─ fileName	string	Yes		SDP file information (The filed name for version compatibility is not displayed temporarily.)	
└─ edidFileName	string	Yes		EDID file name	
└─ reverseControl	object	Yes		Source reverse control information	
└─ ip	string	Yes		Source reverse control IP	
└─ port	integer	Yes		Source reverse control port	
└─ enable	number	Yes		Reverse control switch: 0: Disable. 1: Enable.	
└─ compatibilityM	integer	Yes		EDID compatibility mode: 0: Normal mode (default).	

Name	Type	Required	Default Value	Remarks	More
ode				1: Mac compatibility mode.	
└─ audio	object	Yes		Audio	
└─ isAudioSource	number	Yes		Capability to serve as an audio source: 0: Not supported. 1: Supported.	
└─ audioType	number	Yes		Audio type: -1: No audio. 0: Embedded audio. 1: Analog audio. 2: Dante audio.	
└─ sampling	number	Yes		Sampling Rate	
└─ presetChannelNum	number	Yes		The number of preset channels set by users	
└─ actualChannelNum	number	Yes		The actual number of channels reported at the base level	
└─ sourceId	number	Yes		The interface ID of actual audio source	
└─ channelInfo	object []	Yes			Item type: object
└─ channelId	number	Yes			
└─ channelIndex	number	Yes		The channel index, starting from 0	

Name	Type	Required	Default Value	Remarks	More
 channelType	number	Yes		Connector channel type: 1: RX 2: TX RX stands for input, and TX stands for output, which are applicable for both ordinary and Dante ports.	
 channelLabel	string	Yes		The channel label	
 connectLabel	string	Yes		The label name of Dante connected to peer	
 isMute	number	Yes		Mute switch	
 testAudioEnable	number	Yes		Test switch	
 state	number	Yes		0: Offline. 1: Online. 2: Abnormal.	
 isValid	number	Yes		0: Unavailable. 1: Available.	
 display	object	Yes		Output content information	
 sourceType	number	Yes		Output source type	
 sourceName	string	Yes		Output source name	
 sourceId	number	Yes		Output source ID	

Name	Type	Required	Default Value	Remarks	More
└─ relation	object	Yes			
└─ type	number	Yes		Relation type: 1: Follow. 2: Copy.	
└─ enable	number	Yes		Relation switch: 0: Disable. 1: Enable.	
└─ destinationId	number	Yes		Associated connector ID	
└─ property	number []	Yes		Enumeration of the properties that need to be correlated between the current connectors. 1: timing	Item type: number
└─		No			
└─ hpd	number	Yes		Hot-plug detection, in milliseconds	
└─ threeD	object	Yes		3D information	
└─ actualSource	object	Yes		Actual 3D source information	
└─ sourceFormat	number	Yes		Video source format: 0: None. 1: Top-and-bottom. 2: Frame sequential. 3: Side-by-side.	
└─ presetSource	object	Yes		Preset 3D source information	

Name	Type	Required	Default Value	Remarks	More
 sourceFormat	integer	Yes		Video source format: 0: None. 1: Top-and-bottom. 2: Frame sequential. 3: Side-by-side. 4: From input (default).	
 eyePriority	integer	Yes		Eye priority: 1: Right eye. 2: Left eye (default).	
 threeDEnable	integer	Yes		3D switch enable: 0: Disable. 1: Enable.	
 rightEyeOffset	integer	Yes		Right eye starting position	
 thirdEmitterEnable	integer	Yes		Enable third-party emitter: 1: Disable. 2: Enable.	
 emitterDelay	string	Yes		Emitter delay	
 linkInfo	object	Yes		Link information	
 isLink	number	Yes		Link source or not: 0: Local source. 1: Link source.	
 shared	number	Yes		Whether the local source is shared: 0: Not shared. 1: Shared. Invalid when isLink value is	

Name	Type	Required	Default Value	Remarks	More
				0.	
└─ sourceInfo	object	Yes		Shared source information, valid only when isLink is a link source.	
└─ identify	string	Yes		The unique identification of a shared device	
└─ sourceId	number	Yes		The unique identification of a shared source	
└─ sourceType	number	Yes		Shared source type: 0: Common source. 1: Mosaic source. 2: PGM source.	
└─ count	integer	Yes		Number of items on the current page	mock: 2
└─ page	integer	Yes		Current page number	mock: 1
└─ totalCount	integer	Yes		Total number of items	mock: 2
└─ totalPage	integer	Yes		Total number of pages	mock: 1

3.2 Set Output Mapping

● Basic Information

Path: /v1/node/interface-location

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
nodeId	number	Yes		Node ID	
enable	number	Yes		Output mapping switch: 0: Disable. 1: Enable.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	number	Yes			
message	string	Yes			
data	object	Yes			

3.3 Set Image Quality

● Basic Information

Path: /v1/interface/image-quality

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		
Authorization		Yes		token

Body

Name	Type	Required	Default Value	Remarks	More
	object	No			Item

Name	Type	Required	Default Value	Remarks	More
	□				type: object
└─ interfaceld	number	Yes		Connector ID	
└─ imageQuality	object	Yes			
└─ imageQualityMode	number	Yes		Image quality mode (image quality preset): 0: Standard mode. 1: Document mode. 2: Conference mode. 3: Video mode.	
└─ eyeCare	number	Yes		Eye comfort mode switch: 0: Disable. 1: Enable.	
└─ hue	number	Yes		Hue	
└─ saturation	number	Yes		Saturation	
└─ gamma	number	Yes		Gamma value in image quality. Gamma value [1.00~4.00]*100, to two decimal places.	
└─ gammaScale	number	Yes		Scale ratio multiplier: 1: No scaling. 100: Scaling down by a factor of 100.	
└─ contrast	object	Yes		Contrast	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	

Name	Type	Required	Default Value	Remarks	More
└─ coefficient	number	Yes		Coefficients	
└─ brightness	object	Yes		Brightness	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ coefficient	number	Yes		Coefficients	
└─ shadow	object	Yes		Shadow	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ W	number	Yes		Black level component	
└─ highLight	object	Yes		Highlight	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ W	number	Yes		White level component	
└─ colorTemperature	object	Yes		Color temperature	
└─ mode	number	Yes		Adjustment mode: 0: Quick mode. 1: Advanced mode.	
└─ quickAdjust	object	Yes		Quick parameter adjustment	
└─ value	number	Yes		Color temperature range: 1700~15000, default value: 6500,	

Name	Type	Required	Default Value	Remarks	More
				step: 100, unit: K	
 advanceAdjust	object	Yes		Advanced parameter adjustment	
 R	number	Yes		R component: 0~255, default: 255, step: 1, unit: none	
 G	number	Yes		G component: 0~255, default: 255, step: 1, unit: none	
 B	number	Yes		B component: 0~255, default: 255, step: 1, unit: none	
 isMonochrome	number	Yes		Whether to enable Monochrome: 0: No. 1: Yes.	
 isInverse	number	Yes		Whether to invert colors: 0: No. 1: Yes.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	number	Yes		Response code	
message	string	Yes		Response message	
data	object []	Yes			Item type: object

4. Screen

4.1 v1

4.1.1 FTB

● **Basic Information**

Path: /v1/screen/selected/ftb

Method: PUT

Description:

System global FTB

● **Request Parameters**

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
ftb	object	Yes			
 enable	number	Yes		FTB switch: [0: Disable, return to normal display; 1: Enable.]	
 time	number	Yes		The duration for the FTB effect, which cannot be zero	

● **Return Data**

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			

4.1.2 Freeze

● Basic Information

Path: /v1/screen/selected/freeze

Method: PUT

Description:

Global screen freeze

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
freeze	number	Yes		Screen freeze switch: [0: Unfreeze; 1: Freeze]	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object []	Yes			Item type: object

4.1.3 Take

● Basic Information

Path: /v1/screen/take

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		
Authorization		Yes		token

Body

Name	Type	Required	Default Value	Remarks	More
	object []	No			Item type: object
 screenId	number	No		The screen ID. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	
 screenGuid	string	No		The screen guid	
 effectSelect	number	Yes		Select the transition effect for the take action. [0: Use system default effect, 1: Use protocol-specified effect]	
 direction	number	Yes		Set the take direction: [0: PVW to PGM, 1: PGM to PVW]	
 switchEffect	object	Yes		The transition effect used by all screens	
 time	number	Yes		The transition duration (unit: ms)	
 type	number	Yes		The transition method: [0:	

Name	Type	Required	Default Value	Remarks	More
				CUT, 1: FADE]	
 screenName	string	No		The screen name. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			

4.1.4 Cut

● Basic Information

Path: /v1/screen/cut

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		
Authorization		Yes		token

Body

Name	Type	Required	Default Value	Remarks	More
	object 	No			Item type: object

Name	Type	Required	Default Value	Remarks	More
 screenId	number	No		The screen ID. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	
 direction	number	Yes		Set the cut direction: [0: PVW to PGM, 1: PGM to PVW]	
 screenName	string	No		The screen name. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			

4.2 v2

4.2.1 Retrieve Screen Information

● Basic Information

Path: /v1/screen/list-detail

Method: GET

Description:

● Request Parameters

Query

Name	Required	Example	Remarks
limit	No		Number of items per page
page	No		Query page number
screenId	No		Screen ID. The absence of this parameter indicates it is not considered.
presetId	No		Preset ID, used together with sceneType. Without this parameter, no query is performed. With this parameter, the screen data for the specified preset is queried.
presetType	No		Preset type. Without this parameter, screen information within the preset will not be queried.
screenType	No		Screen type. The absence of this parameter indicates it is not considered.

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			
└─ list	object []	Yes			Item type: object
└─ screenId	number	Yes		Screen ID, which is unique	
└─ screenIdObj	object	Yes		The obj form of the screen ID	
└─ id	number	Yes		ID	

Name	Type	Required	Default Value	Remarks	More
└─ type	number	Yes		Screen type: 0: Empty screen. 2: Common screen. 4: AUX screen. 8: MVR screen. 16: Input view screen. 32: LED screen.	
└─ general	object	Yes		Screen general properties	
└─ name	string	Yes		Screen name	
└─ area	object	Yes		Region information, geographic location	
└─ areaId	integer	Yes		Region ID	
└─ areaName	string	Yes		Region name	
└─ mosaic	object	Yes		Mosaic information	
└─ type	number	Yes		Mosaic type: 0: Standard rectangle mosaic. 1: Standard irregular mosaic. 2: Any irregular mosaic.	

Name	Type	Required	Default Value	Remarks	More
└─ splice	object	Yes		The mosaic information of the screen output connectors. For an irregular mosaic type, use the default (0,0).	
└─ row	number	Yes		Mosaic rows	
└─ column	number	Yes		Mosaic columns	
└─ window	object	Yes		The screen position information (coordinate system relative to canvas)	
└─ x	number	Yes		The starting position of the screen (circumscribed rectangle) in X direction	
└─ y	number	Yes		The starting position of the screen (circumscribed rectangle) in Y	

Name	Type	Required	Default Value	Remarks	More
				direction	
└─ width	number	Yes		The screen width (circumscribed rectangle of the effective output range on the screen)	
└─ height	number	Yes		The screen height (circumscribed rectangle of the effective output range on the screen)	
└─ outputs	object []	Yes			Item type: object
└─ interfaceId	number	Yes		Connector ID	
└─ interfaceOnline	number	Yes		Status of the card where the connector is located: 0: The card is offline, meaning the card is removed. 1: The card is online and the connector connection is	

Name	Type	Required	Default Value	Remarks	More
				normal. 2: The card is online and the connector connection is abnormal. 】	
└─ name	string	Yes		Connector name	
└─ pos	object	Yes		The absolute coordinates of the connector (coordinate system relative to canvas)	
└─ x	number	Yes		The connector position in X direction	
└─ y	number	Yes		The connector position in Y direction	
└─ width	number	Yes		The connector width	
└─ height	number	Yes		The connector height	
└─ crop	object	Yes		The cropping information of the connector display area (relative to the connector)	

Name	Type	Required	Default Value	Remarks	More
 x	number	Yes		The position of the cropping area in X direction	
 y	number	Yes		The position of the cropping area in Y direction	
 width	number	Yes		The width of the cropping area	
 height	number	Yes		The height of the cropping area	
 rotate	object	Yes		Rotation information	
 enable	string	Yes		Rotation switch enable	
 angle	string	Yes		Angle of rotation	
	object []	Yes		Connector feathering area information	Item type: object
 x	number	Yes		Area starting x-coordinate (with the top-left corner as the reference point within the connector)	

Name	Type	Required	Default Value	Remarks	More
└─ y	number	Yes		Area starting y-coordinate (with the top-left corner as the reference point within the connector)	
└─ width	number	Yes		Area width	
└─ height	number	Yes		Area height	
└─ enable	number	Yes		Area switch	
└─ index	number	Yes		Area index	
└─ feather	object []	Yes		Connector feathering information (must describe all four edges of the connector)	Item type: object
└─ enable	number	Yes		Feathering enable: 0: Disable. 1: Enable.	
└─ type	number	Yes		Connector feathering type: 0: Top edge. 1: Bottom edge. 2: Left edge.	

Name	Type	Required	Default Value	Remarks	More
				3: Right edge.	
 gamma	number	Yes		Connector feathering gamma value, with gamma multiplied by 100. For example, a gamma of 2.2 is represented as 220.	
 width	number	Yes		Connector feathering width	
 areaIndex	number	Yes		Index of the area it belongs to	
 area	object []	Yes		Connector feathering area information	Item type: object
 x	number	Yes		Area starting x-coordinate (with the top-left corner as the reference point within the connector)	
 y	number	Yes		Area starting y-coordinate (with the top-left corner as the	

Name	Type	Required	Default Value	Remarks	More
				reference point within the connector)	
└─ width	number	Yes		Area width	
└─ height	number	Yes		Area height	
└─ enable	number	Yes		Area switch	
└─ index	number	Yes		Area index	
└─ isShowRect	integer	Yes		Whether to show the circumscribed rectangle: 0: Do not show the rectangle. 1: Show the rectangle.	
└─ colorDepth	object	Yes		Bit depth	
└─ preset	number	Yes		Preset bit depth: 8: 8Bit. 10: 10Bit. 12: 12Bit. 16: 16Bit. 255: AUTO.	
└─ actual	number	Yes		Actual bit depth: 8: 8Bit. 10: 10Bit.	

Name	Type	Required	Default Value	Remarks	More
				12: 12Bit. 16: 16Bit.	
└─ colorSpace	object	Yes		Color Space	
└─ preset	number	Yes		Preset color space: 0: RGB 4:4:4. 1: YCbCr 4:2:2. 2: YCbCr 4:4:4. 3: YCbCr 4:2:0. 255: AUTO.	
└─ colorRange	object	Yes		Color range	
└─ preset	number	Yes		Preset color range: 0: LIMIT. 1: FULL. 255: AUTO.	
└─ colorGamut	object	Yes		Color gamut	
└─ preset	number	Yes		Preset color gamut: 0: Rec.601. 1: Rec.709. 2: DCI-P3. 3: Rec.2020. 4: NTSC. 5: aRGB. 255: AUTO.	
└─ dynamicRange	object	Yes		Dynamic range	
└─ preset	number	Yes		Preset dynamic range:	

Name	Type	Required	Default Value	Remarks	More
				0: SDR. 1: HDR10. 2: HLG. 255: AUTO.	
└─ actual	number	Yes		Actual dynamic range: 0: SDR. 1: HDR10. 2: HLG.	
└─ smpteMapping	object	Yes		SMPTE 425M Mapping: 0: Unused. 1: Level A. 2: Level B. Currently, only SDI connectors support Level A/Level B settings; other connectors default to unused.	
└─ preset	number	Yes		SMPTE 425M Mapping: 0: Unused. 1: Level A. 2: Level B. Currently, only SDI connectors support Level A/Level B	

Name	Type	Required	Default Value	Remarks	More
				settings; other connectors default to unused.	
└─ timing	object	Yes		Screen connector reference timing	
└─ interlaced	number	Yes		Interlaced scanning: 0: Interlaced. 1: Progressive.	
└─ refresh	number	Yes		Frame rate (floating-point format)	
└─ horizontal	object	Yes		Horizontal timing	
└─ addTime	number	Yes		Visible width (H active, namely the commonly referred resolution width)	
└─ frontPorch	number	Yes		Front porch	
└─ sync	number	Yes		Sync value	
└─ syncPolarity	number	Yes		Sync polarity: 0: Positive. 1: Negative.	

Name	Type	Required	Default Value	Remarks	More
└─ totalTime	number	Yes		Total width (H total)	
└─ vertical	object	Yes		Vertical timing	
└─ addTime	number	Yes		Visible height (V active, namely the commonly referred resolution height)	
└─ frontPorch	number	Yes		Front porch	
└─ sync	number	Yes		Sync value	
└─ syncPolarity	number	Yes		Sync polarity: 0: Positive. 1: Negative.	
└─ totalTime	number	Yes		Total width (H total)	
└─ standardType	string	Yes		Resolution standard: 0: No standard. 1: DMT 2: CVT 3: CVT-RBV1 4: CVT-RBV2 5: CVT-RBV3 6: CEA 7: SMPTE	
└─ thumbTiming	object	Yes		Preset	

Name	Type	Required	Default Value	Remarks	More
				thumbnail timing	
└─ interlaced	number	Yes		Interlaced scanning: 0: Interlaced. 1: Progressive.	
└─ frame	number	Yes		Frame rate (floating-point format)	mock: 1
└─ width	number	Yes		Width	mock: 1920
└─ height	number	Yes		Height	mock: 960
└─ standardType	string	Yes		Resolution standard: 0: No standard. 1: DMT 2: CVT 3: CVT-RBV1 4: CVT-RBV2 5: CVT-RBV3 6: CEA 7: SMPTE	
└─ timingSelect	object	Yes		Connector's timing type	
└─ select	number	Yes		Timing type: 0: presetThumbTiming.	Enumeration: 0, 1, 2

Name	Type	Required	Default Value	Remarks	More
				1: presetTiming.	
└─ backgroundColor	object	Yes		Screen background color	
└─ R	number	Yes		Red component of screen background color	
└─ G	number	Yes		Green component of screen background color	
└─ B	number	Yes		Blue component of screen background color	
└─ layout	object	Yes		The coordinate when multiple screens are displayed simultaneously on the same canvas (the coordinate is consistently referenced from the canvas)	
└─ x	number	Yes		Screen position x-coordinate	

Name	Type	Required	Default Value	Remarks	More
└─ y	number	Yes		Screen position y-coordinate	
└─ physicalSize	object	Yes		The actual size of the physical LED screen	
└─ dotPitch	number	Yes		The pixel pitch of the physical LED screen	
└─ unit	number	Yes		The unit of the physical LED screen size: 0: Pixel 1: Meter	
└─ width	number	Yes		The width of the physical LED screen, measured according to "unit"	
└─ height	number	Yes		The height of the physical LED screen, measured according to "unit"	
└─ enable	number	Yes		Enable: 0: Disable. 1: Enable.	
└─ pgmEdit	number	Yes		The PGM edit permission flag:	

Name	Type	Required	Default Value	Remarks	More
				0: Editing prohibited. 1: Editing allowed.	
└─ select	number	Yes		The screen selection flag: 0: Not selected. 1: Selected.	
└─ imageQuality	object	Yes			
└─ imageQualityMode	number	Yes		Image quality mode (image quality preset): 0: Standard mode. 1: Document mode. 2: Conference mode. 3: Video mode.	
└─ eyeCare	number	Yes		Eye comfort mode switch: 0: Disable. 1: Enable.	
└─ hue	number	Yes		Hue	
└─ saturation	number	Yes		Saturation	
└─ gamma	number	Yes		Gamma value in image quality. Gamma value	

Name	Type	Required	Default Value	Remarks	More
				[1.00~4.00]*100 , to two decimal places.	
└─ gammaScale	number	Yes		Scale ratio multiplier: 1: No scaling. 100: Scaling down by a factor of 100.	
└─ contrast	object	Yes		Contrast	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ coefficient	number	Yes		Coefficients	
└─ brightness	object	Yes		Brightness	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ coefficient	number	Yes		Coefficients	
└─ shadow	object	Yes		Shadow	

Name	Type	Required	Default Value	Remarks	More
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ W	number	Yes		Black level component	
└─ highLight	object	Yes		Highlight	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ W	number	Yes		White level component	
└─ colorTemperature	object	Yes		Color temperature	
└─ mode	number	Yes		Adjustment mode: 0: Quick mode. 1: Advanced mode.	
└─ quickAdjust	object	Yes		Quick parameter adjustment	
└─ value	number	Yes		Color temperature	

Name	Type	Required	Default Value	Remarks	More
				range: 1700~15000, default value: 6500, step: 100, unit: K	
└─ advanceAdjust	object	Yes		Advanced parameter adjustment	
└─ R	number	Yes		R component: 0~255, default: 255, step: 1, unit: none	
└─ G	number	Yes		G component: 0~255, default: 255, step: 1, unit: none	
└─ B	number	Yes		B component: 0~255, default: 255, step: 1, unit: none	
└─ freeze	number	Yes		Screen freeze switch: 0: Unfreeze. 1: Freeze.	
└─ ftb	object	Yes		Screen FTB control	
└─ enable	number	Yes		FTB enable: 0: Disable, and return to normal display.	

Name	Type	Required	Default Value	Remarks	More
				1: Enable.	
└─ time	number	Yes		FTB hold time: 0: Infinite. Other: Hold time (unit: 0.1s. Hold time = actual value * 10).	
└─ testPattern	object	Yes		Test pattern	
└─ enable	number	Yes		Enable: 0: Disable. 1: Enable.	
└─ type	number	Yes		Test pattern type: 0x0000: Black. 0x0001: Red. 0x0002: Green. 0x0003: Blue. 0x0004: White. 0x0005: Vertical stripes. 0x0006: Horizontal stripes. 0x0007: Checkerboard. 0x0100: Horizontal red gradient. 0x0101: Horizontal green gradient.	

Name	Type	Required	Default Value	Remarks	More
				0x0102: Horizontal blue gradient. 0x0103: Horizontal white gradient. 0x0104: Vertical red gradient. 0x0105: Vertical green gradient. 0x0106: Vertical blue gradient. 0x0107: Vertical white gradient. 0x0200: Horizontal white lines. 0x0201: Vertical white lines. 0x0202: White diagonal left. 0x0203: White diagonal right. 0x0204: Crossed lines. 0x0205: Crossed diagonals. 0x0300: Locate. 0xFFFFE: Custom test pattern. 0xFFFF: Off.	

Name	Type	Required	Default Value	Remarks	More
└─ bright	number	Yes		Brightness	
└─ grid	number	Yes		Its meaning is determined by the field "gridType". If the "gridType" field is absent, it indicates the grid density.	
└─ gridType	number	Yes		Description of grid unit: 0: "grid" indicates grid density, adjustable in 8 levels ranging from 1 to 8, with a default of level 5. 1: "grid" indicates grid spacing in pixels, with a minimum of 1 pixel, a maximum equal to 1/4 of the horizontal maximum pixel count for the	

Name	Type	Required	Default Value	Remarks	More
				configured connector, a step size of 1, and a default of 32.	
└─ gridLineWidth	number	Yes		Grid width, unit: pixel	
└─ speed	number	Yes		Speed	
└─ hdr	object	Yes		Connector HDR properties	
└─ sdr	object	Yes		SDR conversion settings	
└─ gamma	number	Yes		Gamma value, range: 1.00~4.00	
└─ hdr10	object	Yes		HDR10 conversion settings	
└─ peakLuminance	number	Yes		Peak screen brightness, range: 0~10000	
└─ envLuminance	number	Yes		Screen's ambient brightness, range: 0~10000	
└─ pqMode	number	Yes		Select PQ curve: 0: ST2084. 1: ST2086.	

Name	Type	Required	Default Value	Remarks	More
└─ toneMapping	number	Yes		Curve adjustment switch: 0: Disable. 1: Enable.	
└─ maxcll	object	Yes		Peak source brightness	
└─ enable	boolean	Yes		Customization enable: 0: Disable. 1: Enable.	
└─ presetMaxcll	number	Yes		Custom peak source brightness, range: 0~10000	
└─ hlg	object	Yes		HLG conversion settings	
└─ peakLuminance	number	Yes		Peak brightness, range: 100~10000	
└─ envLuminance	number	Yes		Ambient brightness	
└─ hdcp	object	Yes		HDCP settings	
└─ enable	number	Yes		Enable status: 0: Disable. 1: Enable.	
└─ presetType	number	Yes		Preset type: 0: Off. 1: HDCP 1.4.	

Name	Type	Required	Default Value	Remarks	More
				2: HDCP 1.3. 3: HDCP 2.2.	
└─ actualType	number	Yes		Actual type: 0: None. 1: HDCP 1.4. 2: HDCP 1.3. 3: HDCP 2.2.	
└─ genlock	object	Yes		Genlock configuration	
└─ type	number	Yes		Genlock type: 0: Internal sync/Disabled. 1: Sync with the Genlock source. 2: Sync with the input source from the connector. 3: Internal frame rate	
└─ frameRate	integer	Yes		Sync frame rate: Can be overridden when syncLockType is set to internal. For other syncLock types, this value does not need to be provided.	

Name	Type	Required	Default Value	Remarks	More
└─ frameRateScale	integer	Yes		Frame rate ratio	
└─ frameRateMode	integer	Yes		Frame rate mode: 0: x1 mode (default). 1: x2 mode.	
└─ phase	object	Yes		Phase offset	
└─ enable	number	Yes		Sync phase adjustment enable: 0: Disable/off. 1: Enable/on.	
└─ offset	number	Yes		Sync shift setting. Unit: clk. Range: -40000~+40000.	
└─ phaseOffsetType	integer	Yes		Phase offset type: 0: Off. 1: Angle. 2: Fraction. 3: Absolute value.	
└─ angle	integer	Yes		Angle	
└─ angleScale	integer	Yes		Ratio of angle	
└─ percentage	integer	Yes		Fraction	

Name	Type	Required	Default Value	Remarks	More
	r				
└─ percentageScale	integer	Yes		Ratio of fraction	
└─ lineCount	integer	Yes		Number of rows	
└─ pixel	integer	Yes		Pixel	
└─ sourceId	number	Yes		Reference source ID: When type=1, sourceId=nodeId. When type=2, sourceId=interfaceId.	
└─ genlockStatus	object	Yes		Current Genlock status	
└─ status	number	Yes		Current Genlock status: 0: Not enabled. 1: Adjusting. 2: Synchronized successfully. 3: Synchronization failed.	
└─ edge	object	Yes		Edge blending	
└─ enable	number	Yes		Enable status: 0: Disable.	

Name	Type	Required	Default Value	Remarks	More
				1: Enable.	
└─ width	number	Yes		Edge width	
└─ height	number	Yes		Edge height	
└─ threeD	object	Yes		3D	
└─ threeDEnable	integer	Yes		3D switch enable: 0: Disable. 1: Enable.	
└─ sourceFormat	integer	Yes		Video source format: 0: None. 1: Top-and-bottom. 2: Frame sequential. 3: Side-by-side. 4: From input (default).	
└─ rightEyeOffset	integer	Yes		Right eye starting position	
└─ eyePriority	integer	Yes		Eye priority: 1: Right eye. 2: Left eye (default).	
└─ thirdEmitterEnable	integer	Yes		Enable third-party emitter: 1: Disable.	

Name	Type	Required	Default Value	Remarks	More
				2: Enable.	
└─ emitterDelay	integer	Yes		Emitter delay	
└─ audio	object	Yes		PGM audio	
└─ isMute	number	Yes		0: Normal. 1: Mute.	
└─ volume	number	Yes		Volume range: 0~100, unit: %	
└─ source	object []	Yes		Input audio source list	Item type: object
└─ type	number	Yes		Audio type: -1: No audio. 0: Embedded audio. 1: Analog audio. 2: Dante audio. 3: Layer.	
└─ id	number	Yes		Connector ID (The layer sends the layer ID.)	
└─ danteChannelIndex	number []	Yes		Dante channel array	Item type: number
└─		No			
└─ output	object []	Yes		Output connector list	Item type: object
└─ accompanyAudioEnable	number	Yes		Whether to enable the	

Name	Type	Required	Default Value	Remarks	More
				embedded audio: 0: Disable. 1: Enable.	
└─ analogVideo	integer []	Yes		Analog audio	Item type: integer
└─		No		Analog audio connectors	
└─ danteChannelIndex	integer []	Yes		Dante channel array	Item type: integer
└─		No			
└─ pvwAudio	object	Yes		PVW audio	
└─ isMute	number	Yes		0: Normal. 1: Mute.	
└─ volume	number	Yes		Volume range: 0~100, unit: %	
└─ source	object []	Yes		Input audio source list	Item type: object
└─ type	number	Yes		Audio type: -1: No audio. 0: Embedded audio. 1: Analog audio. 2: Dante audio. 3: Layer.	
└─ id	number	Yes		Connector ID (The layer sends the layer ID.)	

Name	Type	Required	Default Value	Remarks	More
 danteChannelIndex	number []	Yes		Dante channel array	Item type: number
		No			
 outlet	object []	Yes		Output connector list	Item type: object
 accompanyAudioEnable	number	Yes		Whether to enable the embedded audio: 0: Disable. 1: Enable.	
 analogVideo	integer []	Yes		Analog audio	Item type: integer
		No		Analog audio connectors	
 danteChannelIndex	integer []	Yes		Dante channel array	Item type: integer
		No			
 presetPoll	object	Yes		Preset playlist playback control	
 enable	number	Yes		Preset playlist playback enable	
 presetGroupId	number	Yes		Control preset playlist playback by preset group ID	
 tbar	object	Yes		T-Bar	

Name	Type	Required	Default Value	Remarks	More
				information	
└─ percent	integer	Yes		T-Bar progress	
└─ novs	object	Yes		Screen novs information	
└─ enable	number	Yes		Screen novs switch: 0: Disable. 1: Enable.	
└─ status	number	Yes		Screen novs status: 0: novs inactive (Output displayed normally). 1: novs active (Output blackout).	
└─ threeDLut	object	Yes		3D LUT information	
└─ fileName	string	Yes		3D LUT file name	
└─ fileTitle	string	Yes		3D LUT file title	
└─ enable	integer	Yes		3D LUT enable	
└─ intensity	integer	Yes		3D LUT strength	
└─ curves	object []	Yes			Item type:

Name	Type	Required	Default Value	Remarks	More
					object
└─ point	object	Yes			
└─ input	integer	Yes		Input coordinates	
└─ output	integer	Yes		Output coordinates	
└─ type	integer	Yes			
└─ selectType	integer	Yes		Selected adjustment point	
└─ enable	string	Yes		Curves enable	
└─ index	string	Yes		Index	
└─ colorReplace	object	Yes		Color Replacement	
└─ colorReplaceEnable	number	Yes		Color Replacement Enable: 0: Disable. 1: Enable.	
└─ skinProtectEnable	number	Yes		Skin Tone Protect enable: 0: Disable. 1: Enable.	
└─ srcColor	object	Yes		Original color	
└─ r	integer	Yes		Red component	

Name	Type	Required	Default Value	Remarks	More
└─ g	integer	Yes		Green component	
└─ b	integer	Yes		Blue component	
└─ dstColor	object	Yes		Target color	
└─ r	integer	Yes		Red component	
└─ g	integer	Yes		Green component	
└─ b	integer	Yes		Blue component	
└─ tolerance	number	Yes		Tolerance	
└─ softness	number	Yes		Softness	
└─ shadow	number	Yes		Shadow strength	
└─ colorCorrect	object	Yes		14Ch Color Correction	
└─ enable	integer	Yes		14Ch Color Correction enable: 0: Disable. 1: Enable.	
└─ colorCorrectParams	object []	Yes			Item type: object
└─ colorCorrectType	integer	Yes		14 colors: 0: Red.	

Name	Type	Required	Default Value	Remarks	More
				1: Orange. 2: Yellow. 3: Lime. 4: Green. 5: Turquoise. 6: Cyan. 7: Cobalt. 8: Blue. 9: Violet. 10: Magenta. 11: Crimson. 12: White. 13: Black.	
└─ rgb	object	Yes		When colorCorrectType12 is 0-11, the colors are valid for HSV and invalid for RGB.	
└─ r	integer	Yes		Red (red component)	
└─ Scale	string	Yes		Red component scaling factor	
└─ g	integer	Yes		Green (green component)	
└─ gScale	string	Yes		Green component scaling factor	
└─ b	integer	Yes		Blue (blue component)	

Name	Type	Required	Default Value	Remarks	More
└─ bScale	string	Yes		Blue component scaling factor	
└─ hsv	object	Yes		When colorCorrectType12 is 12-13, the colors are valid for RGB and invalid for HSV.	
└─ h	number	Yes		Hue	
└─ hScale	string	Yes		Hue component scaling factor	
└─ s	number	Yes		Saturation	
└─ sScale	string	Yes		Saturation component scaling factor	
└─ v	number	Yes		Value (brightness)	
└─ vScale	string	Yes		Value component scaling factor	
└─ dynamicBooster	object	Yes		Dynamic Booster	
└─ enable	integer	Yes		Dynamic Booster enable: 0: Disable. 1: Enable.	
└─ level	string	Yes		Dynamic	

Name	Type	Required	Default Value	Remarks	More
				Booster levels: 0~16	
└─ magicGray	object	Yes		magicGray information	
└─ xbitEnable	integer	Yes		22Bit+ enable: 0: Disable. 1: Enable.	
└─ preciseGrayscaleEnable	integer	Yes		Precise Grayscale enable: 0: Disable. 1: Enable.	
└─ functionList	object	Yes		Screen function list	
└─ isSupportPreciseGrayscale	integer	Yes		Whether Precise Grayscale is supported: 0: Not supported. 1: Supported.	
└─ isSupport22BitPlus	integer	Yes		Whether 22bit+ is supported: 0: Not supported. 1: Supported.	
└─ isSupportDynamicBooster	integer	Yes		Whether Dynamic Booster is supported: 0: Not	

Name	Type	Required	Default Value	Remarks	More
				supported. 1: Supported.	
└─ isSupportGamut	integer	Yes		Whether gamut is supported: 0: Not supported. 1: Supported.	
└─ cabinetOverwriteInfo	object	Yes		Cabinet information override	
└─ overwrite	integer	Yes		0: No. 1: Yes	
└─ moduleRow	integer	Yes			
└─ moduleCol	integer	Yes			
└─ multiModelInfo	object	Yes		Multi-mode information	
└─ currentModelId	integer	Yes			
└─ currentCfgParam	object	Yes			
└─ manufactureName	string	Yes			
└─ cabinetName	string	Yes			
└─ cardModel	string	Yes			
└─ status	string	Yes			
└─ version	string	Yes			
└─ issue	integer	Yes			

Name	Type	Required	Default Value	Remarks	More
	array				
└─ modelInfo	object []	Yes		Mode list	Item type: object
└─ modelId	integer	Yes		Mode ID	
└─ modelName	string	Yes		Mode name	
└─ md5	string	Yes			
└─ netportBackup	object	Yes		Ethernet port backup information	
└─ backupType	integer	Yes		Ethernet port backup type: 0: Backup off. 1: Split backup. 2: Sequential backup. 3: Custom backup.	
└─ enableType	integer	Yes		Ethernet port backup enable type: 0: All Ethernet ports on. 1: Primary Ethernet port off. 2: Backup Ethernet port off.	

Name	Type	Required	Default Value	Remarks	More
				3: All Ethernet ports off.	
└─ netportIds	object []	Yes		Ethernet Port backup list	Item type: object
└─ masterNetportId	integer	Yes		Primary Ethernet port	
└─ backupNetportId	integer	Yes		Backup Ethernet port	
└─ screenBoundRect	object	Yes		The screen's circumscribed rectangle	
└─ width	number	Yes		Screen width	
└─ height	number	Yes		Screen height	
└─ adjustment	object	Yes		Adjustment function	
└─ brightness	object	Yes		Brightness	
└─ ratio	integer	Yes		Brightness range: 0~10000	
└─ ratioScale	integer	Yes		Scale ratio multiplier: 1: No scaling. 10000: Scaling down by a factor of 10000.	
└─ gamma	object	Yes			
└─ value	number	Yes		Gamma value	

Name	Type	Required	Default Value	Remarks	More
	er			range: 100~400	
└─ scale	number	Yes		Scale ratio multiplier: 100	
└─ colorTemperature	number	Yes		Color temperature value of the screen	
└─ syncLock	object	Yes			
└─ interfaceID	string	Yes			
└─ syncLockType	integer	Yes		Sync type: 0: Follow input. 1: Genlock. 2: Internal.	
└─ frameRate	number	Yes		Frame rate	
└─ syncLockStatus	integer	Yes		Sync status: 0: Unlocked. 1: Locked.	
└─ phaseOffset	object	Yes			
└─ phaseOffsetType	integer	Yes		Phase offset type: 0: Off. 1: Angle. 2: Fraction. 3: Absolute value.	
└─ angle	number	Yes		Angle, range: 0~342	

Name	Type	Required	Default Value	Remarks	More
└─ percentage	number	Yes		Fraction, range: 0~95	
└─ lineCount	integer	Yes			
└─ pixel	integer	Yes			
└─ frameMultiplication	object	Yes			
└─ frameMultiplicationEnable	integer	Yes			
└─ frameMultiplicationQuantity	integer	Yes			
└─ outputFrames	object []	Yes			Item type: object
└─ frameIndex	integer	Yes			
└─ frameType	integer	Yes			
└─ offset	object	Yes			
└─ x	integer	Yes			
└─ y	integer	Yes			
└─ color	object	Yes			
└─ R	integer	Yes			

Name	Type	Required	Default Value	Remarks	More
└─ G	integer	Yes			
└─ B	integer	Yes			
└─ additionalFrameLatencyValue	integer	Yes		Additional Frame Latency: 0: Off. 1: One frame. 2: Two frames.	
└─ layerLayout	integer	Yes		Layer layout	
└─ screenWorkMode	integer	Yes		Screen working mode	
└─ lowDelayEnable	integer	Yes		Low Latency switch	
└─ lockState	integer	Yes		Screen lock status: 0: Unlocked. 1: Locked.	
└─ interfaceLocation	number	Yes		Output mapping: 0: Disable. 1: Enable.	
└─ enable	number	Yes		Screen switch: 0: Disable. 1: Enable.	
└─ keyPosition	integer []	Yes		The key position index, starting from 0	Item type: integer
└─		No		The position	

Name	Type	Required	Default Value	Remarks	More
				sequence value	
└─ state	integer	Yes		Whether the current screen is blocked: 0: Not blocked. 1: Blocked.	
└─ order	integer	Yes		Screen order, used to record the display order of screens, with a minimum value of 1	
└─ template	object	Yes		The layer template used by the screen	
└─ id	number	Yes		Layer template ID	
└─ moduleType	number	Yes		Layer template module: 0: MVR. 1: Layer.	
└─ type	number	Yes		Template type: 0: Built-in template. 1: Custom template.	
└─ customGamut	object	Yes			
└─ name	string	Yes			
└─ colorGamutInfo	object	Yes			

Name	Type	Required	Default Value	Remarks	More
└─ rLxy	object	Yes			
└─ gamutType	integer	Yes			
└─ lum	integer	Yes			
└─ x	integer	Yes			
└─ y	integer	Yes			
└─ gLxy	object	Yes			
└─ gamutType	integer	Yes			
└─ lum	integer	Yes			
└─ x	integer	Yes			
└─ y	integer	Yes			
└─ bLxy	object	Yes			
└─ gamutType	integer	Yes			
└─ lum	integer	Yes			
└─ x	integer	Yes			
└─ y	integer	Yes			

Name	Type	Required	Default Value	Remarks	More
└─ wLxy	object	Yes			
└─ gamutType	integer	Yes			
└─ lum	integer	Yes			
└─ x	integer	Yes			
└─ y	integer	Yes			
└─ colorTemperature	integer	Yes			
└─ scale	integer	Yes			
└─ originGamut	object	Yes			
└─ rLxy	object	Yes			
└─ gamutType	integer	Yes			
└─ lum	integer	Yes			
└─ x	integer	Yes			
└─ y	integer	Yes			
└─ gLxy	object	Yes			
└─ gamutType	integer	Yes			
└─ lum	integer	Yes			

Name	Type	Required	Default Value	Remarks	More
	r				
└─ x	integer	Yes			
└─ y	integer	Yes			
└─ bLxy	object	Yes			
└─ gamutType	integer	Yes			
└─ lum	integer	Yes			
└─ x	integer	Yes			
└─ y	integer	Yes			
└─ wLxy	object	Yes			
└─ gamutType	integer	Yes			
└─ lum	integer	Yes			
└─ x	integer	Yes			
└─ y	integer	Yes			
└─ colorTemperature	integer	Yes			
└─ scale	integer	Yes			

Name	Type	Required	Default Value	Remarks	More
└─ selectGamut	object	Yes			
└─ presetGamut	integer	Yes			
└─ actualGamut	integer	Yes			
└─ colorTemperature	integer	Yes			
└─ eotf	object	Yes			
└─ shadowCompensation	integer	Yes			
└─ shadowCompensationCoefficient	integer	Yes			
└─ ambientLightCompensation	integer	Yes			
└─ clipLevel	integer	Yes			
└─ peakLuminance	integer	Yes		Peak screen brightness	
└─ peakLumOverrideEnable	integer	Yes		Override screen peak brightness switch: 0: Disable. 1: Enable.	
└─ groupOrder	integer	Yes		The screen order within the group	

Name	Type	Required	Default Value	Remarks	More
└─ group	object	Yes		The screen group information	
└─ id	string	Yes		Group ID: uuid; Ungrouped as ""	
└─ order	integer	Yes		The group order, 0 if ungrouped	
└─ name	string	Yes		Group name, "" if ungrouped	
└─ updateTime	integer	Yes		The update timestamp	
└─ edgeFusion	object	Yes		Edge blending	
└─ enable	number	Yes		0: Disable. 1: Enable.	
└─ mode	string	Yes		0: Basic mode. 1: Advanced mode.	
└─ edgeFusionFlag	object	Yes			
└─ enable	number	Yes		Edge marker switch: 0: Disable. 1: Enable.	
└─ color	object	Yes		The marker color	
└─ R	number	Yes		R	
└─ G	number	Yes		G	

Name	Type	Required	Default Value	Remarks	More
└─ B	number	Yes		B	
└─ guid	string	Yes			
└─ combinationType	integer	Yes		Combination type: 0: Default screen. 1: Cross-device screen. 2: Super screen.	
└─ superScreenGUID	string	Yes			
└─ createTime	number	Yes		The screen creation time	
└─ count	integer	Yes		Number of items on the current page	
└─ page	integer	Yes		Current page number	
└─ totalCount	integer	Yes		Total number of items	
└─ totalPage	integer	Yes		Total pages	

4.2.2 Freeze

● Basic Information

Path: /v1/screen/freeze

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
	object []	No			Item type: object
screenId	number	No		The screen ID. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	
screenGuid	string	Yes		The screen guid	
freeze	number	Yes		Screen freeze switch: 0: Unfreeze. 1: Freeze.	
screenName	string	No		The screen name. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	

Name	Type	Required	Default Value	Remarks	More
data	object []	Yes			Item type: object

4.2.3 FTB

● Basic Information

Path: /v1/screen/ftb

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
	object []	No			Item type: object
└─ screenId	number	No		The screen ID. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	
└─ screenGuid	string	No		The screen guid	
└─ screenName	string	No		The screen name. Provide at least one of screenId or screenName; if both are provided, screenId takes	

Name	Type	Required	Default Value	Remarks	More
				precedence.	
ftb	object	Yes		Screen FTB control	
enable	number	Yes		FTB enable: 0: Disable, and return to normal display. 1: Enable.	
time	number	Yes		The duration for the FTB effect	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object []	Yes			Item type: object

4.2.4 Cut

● Basic Information

Path: /v1/screen/cut

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		
Authorization		Yes		token

Body

Name	Type	Required	Default Value	Remarks	More
	object []	No			Item type: object
 screenId	number	No		The screen ID. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	
 screenGuid	string	No		The screen guid	
 direction	number	Yes		Set the cut direction: 0: PVW to PGM. 1: PGM to PVW.	
 screenName	string	No		The screen name. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	
 swapEnable	number	Yes		Swap switch: 0: Disable. 1: Enable.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			

4.2.4 Take

● Basic Information

Path: /v1/screen/take

Method: PUT

Description:

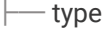
● **Request Parameters**

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		
Authorization		Yes		token

Body

Name	Type	Required	Default Value	Remarks	More
	object []	No			Item type: object
 screenId	number	No		The screen ID. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	
 screenGuid	string	No		The screen guid	
 effectSelect	number	Yes		Select the transition effect for the Take action: 0: Use system default effect. 1: Use protocol-specified effect.	
 direction	number	Yes		Set the Take direction: 0: PVW to PGM. 1: PGM to PVW.	
 switchEffect	object	Yes		The transition effect used by all screens	

Name	Type	Required	Default Value	Remarks	More
 time	number	Yes		The transition duration (unit: ms)	
 type	number	Yes		Transition method: 0: CUT. 1: FADE.	
 swapEnable	number	Yes		Swap switch: 0: Disable. 1: Enable.	
 screenName	string	No		The screen name. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			

5. Layer

5.1 Retrieve Layer Information

- **Basic Information**

Path: /v1/layers/list-detail

Method: GET

Description:

- **Request Parameters**

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Query

Name	Required	Example	Remarks
limit	No		The maximum number of items per page
page	No		Query page number
layerId	No		The unique ID of a layer. If this field is absent, it indicates the IDs of all layers.
id	No		Specify the layer ID under the layer type. If this field is absent, it is ignored.
type	No		Specified the layer type. If this field is absent, it is ignored.
attachScreenId	No		The ID of the screen to which the layer is attached. If this field is absent, it is ignored.
attachScreenType	No		The type of screen to which the layer is attached. If this field is absent, it is ignored.
sceneType	No		The layer type. If this field is absent, it is ignored.

Body

Name	Type	Required	Default Value	Remarks	More
	string	No			format: binary

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Status code	mock: @integer
message	string	Yes		Result information	mock: @string
data	object	Yes		Layer data	
└─ list	object []	Yes			Item type: object
└─ layerId	integer	Yes		The global layer ID, which is unique	
└─ layerIdObj	object	Yes		The layer identifier	
└─ id	integer	Yes		The layer ID	mock: @integer
└─ type	integer	Yes		Layer types: 2: Common layer. 4: AUX layer. 8: MVR layer. 16: BKG layer. 32: Logo layer. 64: Input view layer. 128: Cut layer. 256: Fill layer.	mock: @integer

Name	Type	Required	Default Value	Remarks	More
				512: OSD layer. 1024: Upstream key layer. 2048: Downstream key layer. 4096: Transition key layer. 8192: BKGD layer.	
└─ sceneType	integer	Yes		Where the layer belongs: 1: Not belonging to PGM or PVW. 2: Belonging to PGM. 4: Belonging to PVW. 8: Belonging to both PGM and PVW.	mock: @integer
└─ attachScreenId	integer	Yes		The IDs for different types of the screens: attachScreenType=0, it indicates screenId. attachScreenType=1, it indicates ledScreenId.	mock: @integer
└─ attachScreenType	integer	Yes		Screen type: 0: Screen with video connector. 1: LED screen.	mock: @integer
└─ screenGuid	string	No		The screen GUID. In versions prior to the super screen, this field is empty.	
└─ enable	number	Yes		Layer switch: 0: Disable. 1: Enable.	
└─ selected	integer	Yes		Layer selection: 0: Not selected.	

Name	Type	Required	Default Value	Remarks	More
				1: Selected. Note: The "selected" status includes two functions. If the JSON issued by the software includes only one selected layer, the selection box will blink in the Multiviewer window. If the JSON includes multiple selected layers, the selection box will not blink in the Multiviewer window.	
└─ zorder	number	Yes		Z order	
└─ covered	number	Yes		Layer disabled status: 0: Not disabled. 1: Disabled.	
└─ isLock	number	Yes		Layer lock status: 0: Unlocked. 1: Locked.	
└─ opacity	number	Yes		Layer opacity information, in percentage. For example, 100 represents 100% opacity.	
└─ threeD	object	Yes		Layer 3D related	
└─ enable	number	Yes		The layer 3D switch: 0: Disable. 1: Enable.	
└─ aspectRatio	object	Yes		Layer aspect ratio	

Name	Type	Required	Default Value	Remarks	More
				properties	
└─ aspectRatio	number	Yes		Layer aspect ratio: 0: 1:1 1: 4:3 2: 3:2 3: 16:9 4: 16:10 5: Custom 6: Obsolete 7: 5:4 8: Source ratio	
└─ locked	number	Yes		Lock status: 0: Unlocked. 1: Locked.	
└─ scale	object	Yes		The scaling information	
└─ mode	integer	Yes		Scaling mode: 0: Custom. 1: Pixel-to-pixel. 2: Fullscreen. 3: Follow canvas.	
└─ general	object	Yes		Layer basic Information	
└─ name	string	Yes		Layer name	mock: @string
└─ isBackground	integer	Yes		Whether it is a screen background: 0: No. 1: Yes.	mock: @integer
└─ isFreeze	integer	Yes		Whether the layer is frozen:	mock: @integer

Name	Type	Required	Default Value	Remarks	More
				0: No. 1: Yes.	er
└─ isMonochrome	integer	Yes		Whether it is a monochrome layer: 0: No. 1: Yes.	mock: @integer
└─ isInverse	integer	Yes		Whether it is a color-inverted layer: 0: No. 1: Yes.	mock: @integer
└─ flipType	integer	Yes		Flip type: 0: Normal. 1: Flip horizontally. 2: Flip vertically. 3: Flip horizontally and vertically.	mock: @integer
└─ capacity	integer	Yes		Layer capacity status: 0: SL. 1: DL. 2: 4K. 65535: Auto.	
└─ resource	number	Yes		Layer resource utilization status: 0: No resource utilization, resources not allocated. 1: Resource required, resources allocated. 2: Resource required, resources not allocated. 3: No resource utilization, quantity exceeded.	

Name	Type	Required	Default Value	Remarks	More
└─ interfaces	object []	Yes		The list of connectors exceeding resource limits. Valid when the resource utilization status is 2.	Item type: object
└─ interfaceId	integer	Yes		Connector ID	
└─ source	object	Yes		Layer source	
└─ general	object	Yes		General information	
└─ sourceId	integer	Yes		Source ID, which is unique for a specific source type	mock: @integer
└─ sourceType	integer	Yes		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 9: Mosaic source. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording.	mock: @integer

Name	Type	Required	Default Value	Remarks	More
				19. Cleanfeed.	
└─ connectorType	integer	Yes		Connector type (valid when the source is from an input connector): 0: INVALID 1: EXP 2: Single Link DVI 3: Dual Link DVI 4: HDMI 1.3 5: HDMI 1.4 6: HDMI 2.0 7: Single Link DP 1.1 8: DP 1.2 9: 3G-SDI 10: VGA 11: CVBS 12: YPbPr 13: 12G-SDI 14: 6G-SDI 15: USB 16: HDBaseT 17: Dual Link DP 1.1 18: DVI_Fiber 19: DVI 1.4	
└─ sourceName	string	Yes		Source name	
└─ connectCapacityNum	integer	Yes		Source capacity: [0: SL 1: DL 2: 4K 255: Custom	
└─ relationId	integer	Yes		Source properties related ID. Currently, this field is only meaningful for IPC sources, representing the decodeId.	mock: @integer
└─ primary	object	Yes		Th saved primary source information after the primary source is switched to the backup one	
└─ sourceId	integer	Yes		Source ID, which is unique	

Name	Type	Required	Default Value	Remarks	More
	integer			for a specific source type	
└─ sourceType	integer	Yes		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 9: Mosaic source. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording.	
└─ actual	object	Yes		Th actual displayed layer source	
└─ sourceId	integer	Yes		Source ID, which is unique for a specific source type	
└─ sourceType	integer	Yes		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image.	

Name	Type	Required	Default Value	Remarks	More
				6: LOGO image. 7: IPC type. 8: Cropped source type. 9: Mosaic source. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19: Cleanfeed.	
└─ connectorType	integer	Yes		Connector type (valid when the source is from an input connector): 0: INVALID 1: EXP 2: Single Link DVI 3: Dual Link DVI 4: HDMI 1.3 5: HDMI 1.4 6: HDMI 2.0 7: Single Link DP 1.1 8: DP 1.2 9: 3G-SDI 10: VGA 11: CVBS 12: YPbPr 13: 12G-SDI 14: 6G-SDI 15: USB 16: HDBaseT 17: Dual Link DP 1.1 18: DVI_Fiber 19: DVI 1.4	
└─ sourceName	string	Yes		Source name	
└─ connectCapacityNum	integer	Yes		Source capacity: [0: SL 1: DL 2: 4K	

Name	Type	Required	Default Value	Remarks	More
				255: Custom	
└─ relationId	integer	Yes		Source properties related ID. Currently, this field is only meaningful for IPC sources, representing the decodeId.	
└─ transitionFollow	object	Yes		Transition follow	
└─ enable	integer	Yes		Transition follow switch	
└─ window	object	Yes		Layer layout information (the coordinate system is consistent with the coordinate system of the connector in the screen)	
└─ width	integer	Yes		Layer width	mock: @integer
└─ height	integer	Yes		Layer height	mock: @integer
└─ x	integer	Yes		H offset	mock: @integer
└─ y	integer	Yes		V offset	mock: @integer
└─ virtualWindow	object	Yes		The virtual layer layout information (Send based	

Name	Type	Required	Default Value	Remarks	More
				on the calculation ratio of virtual pixels, unrelated to the device.)	
└─ width	integer	No		Layer width	
└─ height	integer	No		Layer height	
└─ x	integer	No		Horizontal offset	
└─ y	integer	No		Vertical offset	
└─ mask	object	Yes		Layer mask information	
└─ enable	integer	Yes		Whether to enable mask: 0: Disable. 1: Enable.	mock: @integer
└─ left	integer	Yes		Left mask width, in pixels	mock: @integer
└─ leftRatio	integer	Yes		Left mask ratio: leftRatio = round (left mask width/layer width * 100,000)	
└─ right	integer	Yes		Right mask width, in pixels	mock: @integer
└─ rightRatio	integer	Yes		Right mask ratio: rightRatio = round (right mask width/layer width * 100,000)	

Name	Type	Required	Default Value	Remarks	More
 top	integer	Yes		Top mask width, in pixels	mock: @integer
 topRatio	integer	Yes		Top mask ratio: topRatio = round (top mask width/layer height * 100,000)	
 buttom	integer	Yes		Bottom mask width, in pixels	mock: @integer
 buttomRatio	integer	Yes		Bottom mask ratio: buttomRatio = round (bottom mask width/layer height * 100,000)	
 testColor	object	Yes		The test color information for the layer	
 enable	integer	Yes		Whether to enable the layer test pattern: 0: Disable. 1: Enable.	mock: @integer
 R	integer	Yes		The color component value of the layer test pattern	mock: @integer
 G	integer	Yes		The color component value of the layer test pattern	mock: @integer
 B	integer	Yes		The color component value of the layer test pattern	mock: @integer

Name	Type	Required	Default Value	Remarks	More
└─ imageQuality	object	Yes			
└─ imageQualityMode	number	Yes		Image quality mode (image quality preset): 0: Standard mode. 1: Document mode. 2: Conference mode. 3: Video mode.	
└─ eyeCare	number	Yes		Eye comfort mode switch: 0: Disable. 1: Enable.	
└─ hue	number	Yes		Hue	
└─ saturation	number	Yes		Saturation	
└─ gamma	number	Yes		The image quality gamma value, a floating-point number, range: 1.00~4.00	
└─ contrast	object	Yes		Contrast	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ coefficient	number	Yes		Coefficients	
└─ brightness	object	Yes		Brightness	
└─ R	number	Yes		Red component	

Name	Type	Required	Default Value	Remarks	More
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ coefficient	number	Yes		Coefficients	
└─ shadow	object	Yes		Shadow	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ W	number	Yes		Black level component	
└─ highLight	object	Yes		Highlight	
└─ R	number	Yes		Red component	
└─ G	number	Yes		Green component	
└─ B	number	Yes		Blue component	
└─ W	number	Yes		White level component	
└─ colorTemperature	object	Yes		Color temperature	
└─ mode	number	Yes		Adjustment mode: 0: Quick mode.	

Name	Type	Required	Default Value	Remarks	More
				1: Advanced mode.	
└─ quickAdjust	object	Yes		Quick parameter adjustment	
└─ value	number	Yes		Color temperature range: 1700~15000, default value: 6500, step: 100, unit: K	
└─ advanceAdjust	object	Yes		Advanced parameter adjustment	
└─ R	number	Yes		R component: 0~255, default: 255, step: 1, unit: none	
└─ G	number	Yes		G component: 0~255, default: 255, step: 1, unit: none	
└─ B	number	Yes		B component: 0~255, default: 255, step: 1, unit: none	
└─ shadow	object	Yes		Layer shadow information	
└─ general	object	Yes		General information	
└─ enable	integer	Yes		Whether to enable shadow: 0: Disable. 1: Enable.	mock: @integer
└─ feather	integer	Yes		Feathering	mock: @integer
└─ window	object	Yes		Shadow window information	

Name	Type	Required	Default Value	Remarks	More
└─ width	integer	Yes		Width	mock: @integer
└─ height	integer	Yes		Height	mock: @integer
└─ x	integer	Yes		Horizontal offset	mock: @integer
└─ y	integer	Yes		Vertical offset	mock: @integer
└─ color	object	Yes		Shadow color	
└─ R	integer	Yes		Component value Red component value	mock: @integer
└─ G	integer	Yes		Component value Green component value	mock: @integer
└─ B	integer	Yes		Component value Blue component value	mock: @integer
└─ opacity	integer	Yes		Opacity	mock: @integer
└─ crop	object	Yes		Layer cropping	
└─ enable	integer	Yes		Crop enable:	mock:

Name	Type	Required	Default Value	Remarks	More
	r			0: Disable. 1: Enable.	@integer
└─ widthRatio	integer	Yes		widthRatio = round (Crop width/Horizontal resolution * 100,000)	mock: @integer
└─ heightRatio	integer	Yes		heightRatio = round (Crop height/Vertical resolution * 100,000)	mock: @integer
└─ xRatio	integer	Yes		xRatio = round (x-direction offset/Horizontal resolution * 100,000)	mock: @integer
└─ yRatio	integer	Yes		yRatio = round (y-direction offset/Vertical resolution * 100,000)	mock: @integer
└─ cutout	object	Yes		Layer DSK	
└─ enable	integer	Yes		DSK enable: 0: Disable. 1: Enable.	
└─ cursor	object	Yes		Cursor position	
└─ enable	number	Yes		Show cursor: 0: Disable. 1: Enable.	
└─ pos	object	Yes		Cursor coordinates (using the top-left corner of the visible area as the origin)	
└─ x	integer	Yes		Pick point x-coordinate	mock: @integer

Name	Type	Required	Default Value	Remarks	More
 y	integer	Yes		Pick point y-coordinate	mock: @integer
 color	object	Yes		Picked color	
 R	number	Yes		Red component	
 G	number	Yes		Green component	
 B	number	Yes		Blue component	
 type	integer	Yes		DSK type: 0: Luma key. 1: Chroma key. 2: Smart key.	
 brightness	object	Yes		Luma key	
 threshold	integer	Yes		Clip (brightness threshold), range: 0~100	
 gain	integer	Yes		Brightness gain, range: 0~100	
 keyInverse	integer	Yes		Whether to enable invert colors for the key source: 0: Disable. 1: Enable.	
 foregroundColor	object	Yes		Foreground color configuration	
 enable	integer	Yes		Foreground color enable: 0: Disable. 1: Enable.	
 R	integer	Yes		Red component	

Name	Type	Required	Default Value	Remarks	More
	r				
└─ G	integer	Yes		Green component	
└─ B	integer	Yes		Blue component	
└─ keySource	object	Yes		The Key source information	
└─ followInput	integer	Yes		Whether to follow input: 0: Do not follow. 1: Follow.	
└─ sourceType	integer	Yes		Source type: SourceTypeNone = 0 // Empty layer SourceTypeColor = 1 // No source SourceTypeInput = 2 SourceTypePGM = 3 SourceTypePVW = 4 SourceTypeBKG = 5 SourceTypeLOGO = 6 SourceTypeIPC = 7 SourceTypeCrop = 8 // Cropped source SourceTypeSplit = 9 // Mosaic source SourceTypePCSplit = 10 // IPC mosaic SourceTypeInternal = 11 // Internal source SourceTypeInternalGraph = 12 // Internal shape	

Name	Type	Required	Default Value	Remarks	More
				source	
└─ sourceId	integer	Yes		Source ID	
└─ hue	object	Yes		Chroma key	
└─ color	object	Yes		Key color	
└─ R	integer	Yes		Red component	
└─ G	integer	Yes		Green component	
└─ B	integer	Yes		Blue component	
└─ pos	object	Yes		Cursor coordinates (using the top-left corner of the visible area as the origin)	
└─ x	number	Yes		Pick point x-coordinate	
└─ y	number	Yes		Pick point y-coordinate	
└─ hueClip	integer	Yes		Hue clip, range: 5~hueRamp	
└─ hueRamp	integer	Yes		Hue Ramp, range: 5~180	
└─ saturationClip	integer	Yes		Saturation clip, range: 5~100	
└─ saturationGain	integer	Yes		Gain (softness threshold), range: 5~100	
└─ spill	integer	Yes		Spill, range: 5~100	

Name	Type	Required	Default Value	Remarks	More
└─ shadow	integer	Yes		Shadow, range: 5~100	
└─ highLigh	integer	Yes		Highlight, range: 5~100	
└─ edgeSmooth	integer	Yes		Edge smooth: 0: Disable. 1: Enable.	
└─ intelligence	object	Yes		Smart key	
└─ color	object	Yes		Smart key color	
└─ R	integer	Yes		Red component	
└─ G	integer	Yes		Green component	
└─ B	integer	Yes		Blue component	
└─ pos	object	Yes		Cursor coordinates (using the top-left corner of the visible area as the origin)	
└─ x	integer	Yes		Pick point x-coordinate	
└─ y	integer	Yes		Pick point y-coordinate	
└─ mattingStrength	integer	Yes		Keying strength	
└─ gainAdjust	integer	Yes		Gain adjustment	
└─ border	object	Yes		Layer border information	
└─ enable	integer	Yes		Border enable:	mock:

Name	Type	Required	Default Value	Remarks	More
	r			0: Disable. 1: Enable.	@integer
└─ type	integer	Yes		Border type: 0: Hard-in border. 1: Hard-out border. 2: Hard-in-and-out border. 3: Soft border. 4: Halo border. 5: Halo-in border. 6: Halo-out border.	mock: @integer
└─ size	object	Yes		Border width	
└─ hWidth	integer	Yes		Horizontal width	mock: @integer
└─ vWidth	integer	Yes		Vertical width	mock: @integer
└─ color	object	Yes		Border color	
└─ R	integer	Yes		Component value Red component value	mock: @integer
└─ G	integer	Yes		Component value Green component value	mock: @integer
└─ B	integer	Yes		Component value Blue component value	mock: @integer
└─ opacity	integer	Yes		Opacity	mock:

Name	Type	Required	Default Value	Remarks	More
	array				@integer
└─ UMD	object []	Yes		UMD settings (Members in the array can be zero.)	Item type: object
└─ enable	number	Yes		UMD switch: 0: Disable. 1: Enable.	
└─ area	number	Yes		UMD display region: 0: Top left. 1: Middle left. 2: Bottom left. 3: Top center. 4: Middle center. 5: Bottom center. 6: Top right. 7: Middle right. 8: Bottom right.	
└─ type	number	Yes		UMD display content (bitwise OR relationship): Bit 0: Source name. Bit 1: Source type. Bit 2: Source resolution. Bit 3: Display AUX identifier when the source is used on the AUX screen. Bit 4: Display FTB when the source is a screen and the screen is FTB. Bit 5: Display FREEZE	

Name	Type	Required	Default Value	Remarks	More
				when the source is a screen and is the screen is FREEZE.	
└─ name	string	Yes		UMD display content (setting operation, name not relevant)	
└─ color	object	Yes		UMD color information	
└─ foregroundColor	object	Yes		Foreground color	
└─ R	integer	Yes		Red component value	
└─ G	integer	Yes		Green component value	
└─ B	integer	Yes		Blue component value	
└─ opacity	integer	Yes		Opacity percentage (0: fully transparent; 100: fully opaque)	
└─ backgroundColor	object	Yes		Background color	
└─ R	integer	Yes		Red component value	
└─ G	integer	Yes		Green component value	
└─ B	integer	Yes		Blue component value	
└─ opacity	integer	Yes		Opacity percentage (0: fully transparent; 100: fully opaque)	

Name	Type	Required	Default Value	Remarks	More
└─ audio	object	Yes		Layer audio	
└─ enable	integer	Yes		Enable: 0: Disable. 1: Enable.	
└─ volume	integer	Yes		Volume	
└─ isMute	integer	Yes		Mute status: 0: Not muted. 1: Muted.	
└─ qualityEnhance	object	Yes		Layer image quality enhancement	
└─ enable	integer	Yes		Layer image quality enhancement switch: 0: Disable. 1: Enable.	
└─ mode	number	Yes		Image quality enhancement mode: 0: Rich color. 1: High contrast.	
└─ level	number	Yes		Image quality enhancement level: 0: Low. 1: Medium. 2: High.	
└─ cutFill	object	Yes		Cut & Fill properties	
└─ enable	number	Yes		The switch: 0: Disable. 1: Enable.	
└─ curve	object	Yes		Curves	

Name	Type	Required	Default Value	Remarks	More
└─ start	object	Yes		Starting coordinates	
└─ x	integer	Yes		x-coordinate	
└─ y	integer	Yes		y-coordinate	
└─ end	object	Yes		End coordinates	
└─ x	integer	Yes		x-coordinate	
└─ y	integer	Yes		y-coordinate	
└─ relative	object	Yes		Relative coordinates of the Cut layer	
└─ x	integer	Yes		x-coordinate	
└─ y	integer	Yes		y-coordinate	
└─ width	integer	Yes		Width	
└─ height	integer	Yes		Height	
└─ keyPosition	integer []	Yes		The key position index, starting from 0	Item type: integer
└─		No		Position sequence value	
└─ fillId	integer	Yes		The Fill layer ID corresponding to the Cut layer	
└─ cutId	integer	Yes		The Cut layer ID	

Name	Type	Required	Default Value	Remarks	More
	number			corresponding to the Fill layer	
└─ osdId	number	Yes		OSD ID	
└─ screenMode	integer	Yes		Screen working mode: 0: All-In-One Controller. 1: Send-Only Controller.	
└─ serial	number	Yes		Layer number	
└─ keyFrame	object	Yes		KeyFrame properties	
└─ enable	number	Yes		The switch: 0: Disable. 1: Enable.	
└─ speedType	number	Yes		KeyFrame curve speed type: 0: S-curve. 1: Logarithmic curve. 2: Exponential curve. 3: Linear curve.	
└─ startTime	number	Yes		Start time (relative to the Take time as a percentage, coefficient: 1000)	
└─ animationTime	number	Yes		Animation duration (relative to the Take time as a percentage, coefficient: 1000)	
└─ safeFrames	object []	Yes		Safe area information	Item type: object

Name	Type	Required	Default Value	Remarks	More
└─ aspectRatio	number	Yes		Safe area aspect ratio enumeration: 1. 16:9. 2. 4:3. 3. 9:16.	
└─ id	number	Yes		Safe area ID	
└─ osd	object	Yes			
└─ osdType	number	Yes		OSD type: 1: Text. 2: Image.	
└─ preset	object	Yes		Preset information	
└─ isUsed	number	Yes		Whether to enable preset	
└─ id	number	Yes		The used preset ID, effective when isUsed=1	
└─ osdInfo	object	Yes		OSD layer information	
└─ text	object	Yes		Required when osdType=1	
└─ components	object []	Yes		Widget list	Item type: object
└─ background	object	Yes		Background properties	
└─ color	object	Yes		Background color	
└─ Alpha	number	Yes		Transparency	
└─ R	number	Yes		Color component - R	

Name	Type	Required	Default Value	Remarks	More
└─ G	number	Yes		Color component - G	
└─ B	number	Yes		Color component - B	
└─ enable	number	Yes		Background switch: 0: Disable. 1: Enable.	
└─ font	object	Yes		Text properties	
└─ color	object	Yes		Text color	
└─ Alpha	number	Yes		Transparency	
└─ R	number	Yes		Color component - R	
└─ G	number	Yes		Color component - G	
└─ B	number	Yes		Color component - B	
└─ stroke	object	No		Stroke	
└─ color	object	Yes		Stroke color	
└─ Alpha	number	Yes		Transparency	
└─ R	number	Yes		Color component - R	
└─ G	number	Yes		Color component - G	
└─ B	number	Yes		Color component - B	
└─ enable	number	Yes		Background switch:	

Name	Type	Required	Default Value	Remarks	More
	er			0: Disable. 1: Enable.	
└─ language	number	Yes		0: Simplified Chinese. 1: English.	
└─ componentId	string	Yes		Widget ID	
└─ componentType	number	Yes		Widget type: 0: Static text. 1: Dynamic text. 3: Time. 4: Weather.	
└─ fontId	string	Yes		Font ID	
└─ fontSize	number	Yes		Font size	
└─ wordSpace	number	Yes		Spacing	
└─ lineSpace	number	Yes		Line spacing	
└─ italic	number	Yes		Italic switch	
└─ bold	number	Yes		Bold switch	
└─ underline	number	Yes		Underline switch	
└─ horizontalAlign	number	Yes		Horizontal alignment: 0: Align left. 1: Center. 2: Align right.	
└─ verticalAlign	number	Yes		Vertical alignment:	

Name	Type	Required	Default Value	Remarks	More
	enum			0: Align top. 1: Center. 2: Align bottom.	
└─ position	object	Yes		Widget position information	
└─ x	number	Yes		Position: x-coordinate	
└─ y	number	Yes		Position: y-coordinate	
└─ width	number	Yes		Position: width	
└─ height	number	Yes		Position: height	
└─ staticText	object	Yes		Static text	
└─ direction	number	Yes		Text direction: 0: Left to right. 2: Top to bottom.	
└─ word	string	Yes		Text content. Use “\n” to separate multiple lines.	
└─ dynamicText	object	Yes		Dynamic text	
└─ direction	number	Yes		Text direction: 0: Left to right. 1: Right to left.	
└─ speed	number	Yes		Speed	
└─ word	string	Yes		Text content. Use “\n” to separate multiple lines.	
└─ date	object	Yes		Date	

Name	Type	Required	Default Value	Remarks	More
 timeZone	number	Yes		Timezone identifier, range 0~89. See node configuration file definition.	
 timeType	number	Yes		Time type: 0: Local time. 1: NTP time.	
 timeOffset	number	Yes		Time offset	
 isMultiLine	number	Yes		Whether to display in multiple lines	
 componentSpace	number	Yes		Widget spacing	
 yearFormat	number	Yes		Year format: 0: Four-digit year (YYYY). 1: Two-digit year (YY).	
 dateFormat	number	Yes		Date format: 0: YYYY/MM/DD 1: YYYY/M/D 2: MM/DD/YYYY 3: M/D/YYYY 4: DD/MM/YYYY 5: D/M/YYYY 6: YYYY-MM-DD 7: YYYY-M-D 8: MM-DD-YYYY 9: M-D-YYYY 10: DD-MM-YYYY 11: D-M-YYYY 12: YYYY(year)MM(month)DD	

Name	Type	Required	Default Value	Remarks	More
				(day) 13: YYYY(year)M(month)D(day)	
└─ isDisplayWeek	number	Yes		Whether to display the day of the week	
└─ hourly	number	Yes		Hour system: 0: 12-hour. 1: 24-hour.	
└─ timeFormat	number	Yes		0: HH:MM:SS (default) 1: HH:MM 2: H:M:S 3: H:M 4: H(hour)M(minute)S(second) 5: H(hour)M(minute)	
└─ isDisplayAm	number	Yes		Whether to display AM/PM	
└─ isDisplayDate	number	Yes		Whether to display the date	
└─ isDisplayTime	number	Yes		Whether to display the time	
└─ url	string	Yes		NTP address	
└─ weather	object	Yes		Weather	
└─ location	string	Yes		Location information, with latitude and longitude separated by a comma, in the format: longitude, latitude	

Name	Type	Required	Default Value	Remarks	More
 refreshTime	number	Yes		Refresh interval, in seconds	
 isMultiLine	number	Yes		Whether to display in multiple lines	
 componentSpace	number	Yes		Widget spacing	
 temperatureUnit	number	Yes		Temperature unit: 0: Celsius (C). 1: Fahrenheit (F).	
 weather	object	Yes		Weather	
 enable	number	Yes		Whether to display	
 label	string	Yes		Weather label	
 value	string	Yes		Weather information	
 temp	object	Yes		Temperature	
 enable	number	Yes		Whether to display	
 label	string	Yes		Temperature label	
 value	string	Yes		Temperature information	
 tempNow	object	Yes		Current temperature	
 enable	number	Yes		Whether to display	
 label	string	Yes		Current temperature label	
 value	string	Yes		Current temperature information	
 wind	object	Yes		Wind	
 enable	number	Yes		Whether to display	

Name	Type	Required	Default Value	Remarks	More
	er				
└─ label	string	Yes		Wind speed label	
└─ value	string	Yes		Wind speed information	
└─ humidity	object	Yes		Humidity	
└─ enable	number	Yes		Whether to display	
└─ label	string	Yes		Humidity label	
└─ value	string	Yes		Humidity information	
└─ picture	object	Yes		Required when osdType=2	
└─ pictureId	number	Yes		Image ID	
└─ componentId	string	Yes		The widget's unique ID	
└─ opacity	number	Yes		Opacity	
└─ position	object	Yes		Position	
└─ x	number	Yes		Position: x-coordinate	
└─ y	number	Yes		Position: y-coordinate	
└─ width	number	Yes		Position: width	
└─ height	number	Yes		Position: height	
└─ count	integer	Yes		Number of items on the current page	

Name	Type	Required	Default Value	Remarks	More
└─ page	integer	Yes		Current page number	
└─ totalCount	integer	Yes		Total number of items	
└─ totalPage	integer	Yes		Total pages	

5.2 Retrieve Layer Templates

- **Basic Information**

Path: /v1/layers/template

Method: GET

Description:

- **Request Parameters**

Query

Name	Required	Example	Remarks
moduleType	No	1	Data associated function module: 0: Multiviewer. 1: Layer.
templateId	No	1	Template ID. If not provided, query based on type.

- **Return Data**

Name	Type	Required	Default Value	Remarks	More
code	number	Yes		0: Normal. 8212: Business exception.	
message	number	Yes			
data	object []	Yes			Item type:

Name	Type	Required	Default Value	Remarks	More
					object
└─ layout	object []	Yes		Template layout information	Item type: object
└─ row	integer	Yes			
└─ column	integer	Yes			
└─ type	integer	Yes		Layout type: 1: All. 2: Output. 3: Input.	
└─ isHide	integer	Yes		0: Show. 1: Hide.	
└─ margin	object	Yes		Border information	
└─ top	integer	Yes		Outer border - top	
└─ bottom	string	Yes		Outer border - bottom	
└─ left	string	Yes		Outer border - left	
└─ right	string	Yes		Outer border - right	
└─ cells	object []	Yes		Grid information	Item type: object
└─ row	number	Yes		Grid row - position	
└─ column	number	Yes		Grid column - position	
└─ rowSpan	number	Yes		Grid row width	
└─ columnSpan	number	Yes		Grid column width	
└─ sourceType	number	Yes		Source type: 0: Empty layer.	

Name	Type	Required	Default Value	Remarks	More
				1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 9: Mosaic source. 10: IPC mosaic screen. 11: 11: Internal source. 12: Internal shape source.	
└─ margin	object	Yes		Border information	
└─ top	integer	Yes		External border - top	
└─ bottom	string	Yes		Outer border - bottom	
└─ left	string	Yes		Outer border - left	
└─ right	string	Yes		Outer border - right	
└─ templateId	number	Yes		Template unique ID (original field)	
└─ templateIdObj	object	Yes		Template ID OBJ (original field)	
└─ id	number	Yes		Template ID	
└─ type	number	Yes		Template type: 1: Built-in template. 2: Custom	

Name	Type	Required	Default Value	Remarks	More
				template. 3: Aging test template.	
└─ moduleType	number	Yes		Associated module: 0: Multiviewer. 1: Layer.	
└─ combinationData	object []	Yes		Table combination, used for layer layout	Item type: object
└─ row	integer	Yes		Grid row - position	
└─ column	integer	Yes		Grid column - position	
└─ rowSpan	integer	Yes		Grid row width	
└─ columnSpan	integer	Yes		Grid column width	
└─ radius	object []	Yes		Layer movement range	Item type: object
└─ sourceType	number	Yes		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 9: Mosaic source. 10: IPC mosaic	

Name	Type	Required	Default Value	Remarks	More
				screen. 11: 11: Internal source. 12: Internal shape source.	
└─ minX	number	Yes		Minimum movement range value x	
└─ minY	number	Yes		Minimum movement range value y	
└─ maxX	number	Yes		Maximum movement range value x	
└─ maxY	number	Yes		Maximum movement range value y	

5.3 Apply Layer Templates

● Basic Information

Path: v1/layers/template/select

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
templateId	number	Yes		Used template unique ID	
screenId	number	Yes		Unique ID of the screen where the template is applied	
sceneType	number	Yes		scenetype of the screen where the template is applied, same as the sceneType field definition in the layer	
moduleType	number	No		Associated module: 0: Multiviewer. 1: Layer. (Used from version 1.4 onwards. Not provided in older versions; does not affect functionality.)	
templateType	number	No		Template type: 1: Built-in template. 2: Custom template. 3: Aging test template. (Used from version 1.4 onwards. Not provided in older versions; does not affect functionality.)	
locked	number	Yes		Layer aspect ratio lock status: 0: Unlocked. 1: Locked.	

● **Return Data**

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	

Name	Type	Required	Default Value	Remarks	More
message	string	Yes		Response message	
data	object []	Yes		Layer data	Item type: object
└─ templateId	number	Yes		Used template unique ID	
└─ screenId	string	Yes		Unique ID of the screen where the template is applied	
└─ sceneType	string	Yes		scenetype of the screen where the template is applied, same as the sceneType field definition in the layer	
└─ position	object []	Yes			Item type: object
└─ window	object	Yes		Layer layout information (the top-left corner of the screen's circumscribed rectangle as the coordinate origin).	
└─ width	integer	Yes		The layer width The layer width	mock: @integer

Name	Type	Required	Default Value	Remarks	More
└─ height	integer	Yes		The layer height The layer height	mock: @integer
└─ x	integer	Yes		H offset H offset	mock: @integer
└─ y	integer	Yes		V offset V offset	mock: @integer
└─ source	object	Yes		Layer source information	
└─ general	object	Yes		General information	
└─ sourceId	integer	Yes		Source ID, which is unique for a specific source type	mock: @integer
└─ sourceType	integer	Yes		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 9: Mosaic source. 10: IPC mosaic screen. 11: Internal	mock: @integer

Name	Type	Required	Default Value	Remarks	More
				source]	
└─ connectorType	integer	Yes		Connector type (valid when the source is from an input connector): 0: INVALID 1: EXP 2: Single Link DVI 3: Dual Link DVI 4: HDMI 1.3 5: HDMI 1.4 6: HDMI 2.0 7: Single Link DP 1.1 8: DP 1.2 9: 3G-SDI 10: VGA 11: CVBS 12: YPbPr 13: 12G-SDI 14: 6G-SDI 15: USB 16: HDBaseT 17: Dual Link DP 1.1 18: DVI_Fiber 19: DVI 1.4	
└─ sourceName	string	Yes		The source name	
└─ connectCapacityNum	integer	Yes		Layer capacity status:	

Name	Type	Required	Default Value	Remarks	More
				0: SL. 1: DL. 2: 4K. 65535: Auto.	

5.4 Set Layer Source

● Basic Information

Path: /v1/layers/source

Method: PUT

Description:

Internal shape name	Enumeration
Circle	1
Square	2
Rectangle	3
Ellipse	4
Triangle	5
Diamond	6
Heart	7
Cloud	8
Four-pointed star	9
Five-pointed star	10
Invalid	255

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
	object []	No		Layer data	Item type: object
└─ layerId	integer	Yes		The global layer ID, which is unique	
└─ source	object	Yes		Layer source information	
└─ general	object	Yes		General information	
└─ sourceId	integer	Yes		Source ID: The unique ID of the source under the source type (if it's an internal shape source, this field represents the shape enumeration value)	mock: @integer
└─ sourceType	integer	Yes		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 9: Mosaic source. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter.	mock: @integer

Name	Type	Required	Default Value	Remarks	More
				17: Streaming. 18: Recording. 19: Cleanfeed.	
 relationId	integer	Yes		Source properties related ID. Currently, this field is only meaningful for IPC sources, representing the decodeId.	mock: @integer

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code: Definition can be found in <i>NovaStar Unified Platform Device Error Code Definition</i> .	
message	string	Yes		Response message	
data	object []	Yes			Item type: object

5.5 Select Layer

● Basic Information

Path: /v1/layers/select

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
------	-------	----------	---------	---------

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
	object []	No		Layer data	Item type: object
 layerId	integer	Yes		The global layer ID, which is unique	
 selected	integer	Yes		Whether the layer is selected: 0: Not selected. 1: Selected. Note: The "selected" status includes two functions. If the JSON issued by the software includes only one selected layer, the selection box will blink in the Multiviewer window. If the JSON includes multiple selected layers, the selection box will not blink in the Multiviewer window.	mock: @integer

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code: Definition can be found in <i>NovaStar Unified Platform Device Error Code Definition</i> .	
message	string	Yes		Response message	

Name	Type	Required	Default Value	Remarks	More
data	object []	Yes			Item type: object

6. Preset

6.1 v1

6.1.1 Edit Preset

● **Basic Information**

Path: /v1/preset/general

Method: PUT

Description:

● **Request Parameters**

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
	object []	No			Item type: object
└─ presetId	integer	No		The preset ID. Provide at least one of presetId or presetName; if both are provided, presetId takes precedence.	
└─ general	object	Yes		General information	
└─ name	string	Yes		Preset name	
└─ presetName	string	No		The original preset name. Provide at least one of	

Name	Type	Required	Default Value	Remarks	More
				presetId or presetName; if both are provided, presetId takes precedence.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	number	No		Response code	
message	string	No		Response message	
data	object []	No			Item type: object
 presetId	string	Yes		The IDs of the failed presets during batch modification	

6.1.2 Save Preset

● Basic Information

Path: /v1/preset/create-assign

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
namePrefix	string	No		The preset name prefix.	

Name	Type	Required	Default Value	Remarks	More
				Provide at least one of name or namePrefix; if both are provided, name takes precedence.	
presetIdObj	object	Yes		The preset ID information	
└─ id	integer	Yes		The preset number ID	
└─ sceneType	number	Yes		The preset is from PVW or PGM: [2: PGM; 4: PVW)	
keyPosition	integer []	Yes		The key position index, starting from 0	Item type: integer
└─		No		The position sequence value	
screenInfo	object []	Yes		The parameter of the screen in the preset. Fill in this value in independent preset mode.	Item type: object
└─ screenId	integer	No		The ID of the screen in the preset. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	
└─ screenName	string	No		The screen name. Provide at least one of screenId or screenName; if both are provided, screenId takes precedence.	

Name	Type	Required	Default Value	Remarks	More
name	string	No		The preset name. Provide at least one of name or namePrefix; if both are provided, name takes precedence.	mock: Preset name

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	number	No		Response code	
message	string	No		Response message	
data	object []	Yes			Item type: object
└─ presetId	integer	Yes		The preset ID	
└─ presetIdObj	object	Yes		The preset ID information	
└─ id	integer	Yes		The preset number ID	
└─ sceneType	string	Yes		The preset is from PVW or PGM (0: PGM; 1: PVW)	
└─ general	object	Yes		General information (for possible future expansion)	
└─ name	string	Yes		Preset name	
└─ screens	object []	Yes		The screen information	Item type: object
└─ screenId	integer	Yes		The screen ID, which	

Name	Type	Required	Default Value	Remarks	More
				is unique	
└─ screenIdObj	object	Yes		The obj form of the screen ID	
└─ id	number	Yes		ID	
└─ type	number	Yes		Screen type: [2: Common screen; 4: AUX screen; 8: MVR screen; 16: Input view screen]	
└─ general	object	Yes			
└─ name	string	Yes		The screen name	
└─ mosaic	object	Yes		The mosaic information	
└─ type	number	Yes		Mosaic type: [0: Standard rectangle mosaic; 1: Standard irregular mosaic; 2: Any irregular mosaic]	
└─ splice	object	Yes		The mosaic information of the screen output connectors. For an irregular mosaic type, use the default (0,0).	
└─ row	number	Yes		The mosaic rows	
└─ column	number	Yes		The mosaic columns	
└─ window	object	Yes		The screen position	

Name	Type	Required	Default Value	Remarks	More
				information (coordinate system relative to canvas)	
└─ x	number	Yes		The starting position of the screen (circumscribed rectangle) in X direction	
└─ y	number	Yes		The starting position of the screen (circumscribed rectangle) in Y direction	
└─ width	number	Yes		The screen width (circumscribed rectangle of the effective output range on the screen)	
└─ height	number	Yes		The screen height (circumscribed rectangle of the effective output range on the screen)	
└─ outputs	object []	Yes			Item type: object
└─ interfaced	number	Yes		The connector ID	
└─ pos	object	Yes		The absolute coordinates of the connector (coordinate	

Name	Type	Required	Default Value	Remarks	More
				system relative to canvas)	
└─ x	number	Yes		The connector position in X direction	
└─ y	number	Yes		The connector position in Y direction	
└─ width	number	Yes		The connector width	
└─ height	number	Yes		The connector height	
└─ crop	object	Yes		The cropping information of the connector display area (relative to the connector)	
└─ x	number	Yes		The position of the cropping area in X direction	
└─ y	number	Yes		The position of the cropping area in Y direction	
└─ width	number	Yes		The width of the cropping area	
└─ height	number	Yes		The height of the cropping area	
└─ layers	object []	Yes		The layer list	Item type: object
└─ layerId	integer	Yes		The global layer ID, which is unique	

Name	Type	Required	Default Value	Remarks	More
└─ layerIdObj	object	Yes		The layer identifier	
└─ id	integer	Yes		The layer ID	mock: @integer
└─ type	integer	Yes		Layer type: [2: Common layer; 4: AUX layer; 8: MVR layer; 16: BKG layer; 32: Logo layer]	mock: @integer
└─ sceneType	integer	Yes		Where the layer belongs: [1: Not belonging to PGM or PVW; 2: Belonging to PGM; 4: Belonging to PVW]	mock: @integer
└─ attachScreenId	integer	Yes		The IDs for different types of the screens [attachScreenType==0, it indicates screenId; attachScreenType==1, it indicates ledScreenId]	mock: @integer
└─ attachScreenType	integer	Yes		Screen type: [0: Screen with video connectors; 1: LED screen]	mock: @integer
└─ general	object	Yes		The layer basic information	
└─ name	string	Yes		The layer name	mock: @string

Name	Type	Required	Default Value	Remarks	More
└─ source	object	Yes		The layer source	
└─ general	object	Yes		The general Information	
└─ sourceId	integer	Yes		The source ID, which is unique for a specific source type	mock: @integer
└─ sourceType	integer	Yes		Source type: [0: Empty layer; 1: No-source layer; 2: Input type; 3: PGM; 4: PVW; 5: BKG image 6: LOGO image; 7: IPC type; 8: Cropped source type; 9: Mosaic source; 10: IPC mosaic screen; 11: Internal source]	mock: @integer
└─ window	object	Yes		The layer layout information (the top-left corner of the screen's circumscribed rectangle as the coordinate origin).	
└─ width	integer	Yes		The layer width	mock: @integer
└─ height	integer	Yes		The layer height	mock: @integer
└─ x	integer	Yes		The H Offset	mock: @integer

Name	Type	Required	Default Value	Remarks	More
					r
└─ y	integer	Yes		The V Offset	mock: @integer
└─ enable	number	Yes		Layer switch: [1: Enable; 0: Disable]	
└─ keyPosition	integer []	Yes		The key position index, starting from 0	Item type: integer
└─		No		The position sequence value	

6.1.3 Load Preset

● Basic Information

Path: /v1/preset/play

Method: PUT

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
presetId	integer	No		The preset ID. At least one of presetId, id, or presetName should be provided. The priority order when all are present is	

Name	Type	Required	Default Value	Remarks	More
				presetId, id, presetName.	
sceneType	integer	Yes		Where the preset will be played: 2: PGM; 4: PVW; 0: Auto (PVW for switchers, splicers, PGM for controllers)	
id	integer	No		The preset number ID. Default value: -1, normally starts from 0, using the specified preset slot ID when creating the preset. At least one of presetId, id, or presetName should be provided. The priority order when all are present is presetId, id, presetName.	
presetName	string	No		The preset name. At least one of presetId, id, or presetName should be provided. The priority order when all are present is presetId, id, presetName.	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	number	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			
└─ screenObjFailed	object []	Yes			Item type: object

Name	Type	Required	Default Value	Remarks	More
└─ id	number	Yes		ID	
└─ type	number	Yes		Screen type: [2: Common screen; 4: AUX screen; 8: MVR screen; 16: Input view screen]	
└─ screenObjResourceNotEnough	object []	Yes			Item type: object
└─ id	number	Yes		ID	
└─ type	number	Yes		Screen type: [2: Common screen; 4: AUX screen; 8: MVR screen; 16: Input view screen]	

6.2 v2

6.2.1 Retrieve Preset Details

- **Basic Information**

Path: /v1/preset

Method: GET

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Query

Name	Required	Example	Remarks
limit	No		Number of items per page
page	No		Query page number
type	No		Query the specified preset type

Body

Name	Type	Required	Default Value	Remarks	More
screenId	integer	Yes		Screen ID	
presetId	integer	Yes		presetId	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			
└─ list	object []	Yes			Item type: object
└─ guid	string	Yes		Preset ID	
└─ serial	integer	Yes		Preset ID	
└─ name	string	Yes		Preset name	

Name	Type	Required	Default Value	Remarks	More
└─ type	integer	Yes		Preset type: 1: Global. 2: Single screen. 3: Multi-screen. 4: Screen preset (screen template).	
└─ loadType	integer	Yes		Preset loading method: 1: Complete. 2: Relative.	
└─ sourceRegion	integer	Yes		Preset saved from: 2: PWG. 4: PVW.	
└─ switchEffect	object	Yes		The global transition effect	
└─ time	number	Yes		The transition duration (unit: ms)	
└─ type	number	Yes		Transition effects: 0: Cut. 1: Fade. 2. Dip. 3. Wipe. 4. Stinger. 5. DVE.	
└─ wipe	object	No		Wipe	
└─ image	integer	No		Pattern settings: 1. Vertical rectangle: The height remains constant while the width increases from left to right, and decreases in the	

Name	Type	Required	Default Value	Remarks	More
				opposite direction when reversed. The border appears only on the moving edge (middle vertical line). 2. Horizontal rectangle: The width remains constant while the height increases from top to bottom, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle horizontal line). 3. Vertical rectangle: The height remains constant while the width increases from the center towards both sides, and decreases in the opposite direction when reversed. The border appears only on the moving edge (vertical lines moving outward from the center). 4. Horizontal rectangle: The width	

Name	Type	Required	Default Value	Remarks	More
				remains constant while the height increases from top to bottom, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle horizontal line). 5. Center rectangle. 6. Top-left rectangle. 7. Top-right rectangle. 8. Bottom-right rectangle. 9. Bottom-left rectangle. 10. Top-center rectangle. 11. Right-center rectangle. 12. Bottom-center rectangle. 13. Left-center rectangle. 14. Cross shape. 15. Center diamond. 16. Center ellipse. 17. Top-left triangle. 18. Top-right triangle.	
 symmetry	integer	No		Symmetry, ranging from 0% to 100%, with a coefficient of 100	

Name	Type	Required	Default Value	Remarks	More
				(needs to be multiplied by 100)	
└─ posX	integer	No		The X position where the pattern appears	
└─ posY	integer	No		The Y position where the pattern appears	
└─ frameW	integer	No		Border width	
└─ frameS	integer	No		Border softness	
└─ direction	integer	No		Direction: 1. Forward. 2. Reverse. 3. Switch.	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source.	

Name	Type	Required	Default Value	Remarks	More
				12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ dve	object	No		dve	
└─ type	integer	No		Type: Type: 1. Push. 2. Squeeze. 3. Material effect.	
└─ image	integer	No		Image: Based on the UI, from left to right and top to bottom.	
└─ brightness	integer	No		Clip	
└─ softness	integer	No		Softness	
└─ directionDev	integer	No		Dve direction: 0. Upper left 1. Up 2. Upper right 3. Right 4. Lower right 5. Down 6. Lower left 7. Left	
└─ directionEffect	string	No		Effect direction: 1. Forward. 2. Reverse.	

Name	Type	Required	Default Value	Remarks	More
				3. Switch.	
└─ keySource	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ KeySourceReverse	integer	No		Whether to enable invert colors for the	

Name	Type	Required	Default Value	Remarks	More
				key source: 0: Disable. 1: Enable.	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19: Cleanfeed.	

Name	Type	Required	Default Value	Remarks	More
└─ stinger	object	No		stinger	
└─ playStartTime	integer	No		Playback start time	
└─ mixtureTime	integer	No		Mix time	
└─ mixtureStartTime	integer	No		Mix start time	
└─ matting	integer	No		DSK switch: 1: Enable. 0: Disable.	
└─ brightness	integer	No		Clip	
└─ softness	integer	No		Softness	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape	

Name	Type	Required	Default Value	Remarks	More
				source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ dip	object	No		Dip	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source.	

Name	Type	Required	Default Value	Remarks	More
				14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ currentRegion	integer	Yes		Current on-screen area: 0: Not displayed 2: PWG. 4: PVW. 6: PGM & PVW.	
└─ screens	object []	Yes		Screen information	Item type: object
└─ guid	string	Yes		Screen GUID	
└─ screenIdObj	object	Yes		The obj form of the screen ID	
└─ id	number	Yes		Screen ID	
└─ type	number	Yes		Screen type: 0: Empty screen. 2: Common screen. 4: AUX screen. 8: MVR screen. 16: Input view screen. 32: LED screen.	
└─ name	string	Yes		Screen name	
└─ exist	integer	Yes		Screen presence: 0: Not present. 1: Present.	

Name	Type	Required	Default Value	Remarks	More
└─ window	object	Yes		Screen window information	
└─ x	integer	Yes		H offset	
└─ y	integer	Yes		V offset	
└─ width	integer	Yes		Width	
└─ height	integer	Yes		Height	
└─ layers	object []	Yes		Layer information	Item type: object
└─ serial	number	Yes		Layer number	
└─ type	number	Yes		Layer type	
└─ window	object	Yes		Layer position information	
└─ x	integer	Yes		H offset	
└─ y	integer	Yes		V offset	
└─ width	integer	Yes		Layer width	
└─ height	integer	Yes		Layer height	
└─ enable	integer	Yes		Layer switch	
└─ osd	object	Yes		OSD information	
└─ type	number	Yes		OSD type: 1: Text. 2: Image.	
└─ id	number	Yes		Unique OSD identification Preset ID (The type is 1.) Image ID (The type is 2.)	
└─ count	integer	Yes		Number of items on	

Name	Type	Required	Default Value	Remarks	More
				the current page	
└─ page	integer	Yes		Current page number	
└─ totalCount	integer	Yes		Total number of items	
└─ totalPage	integer	Yes		Total pages	

6.2.2 Create Preset

● Basic Information

Path: /v1/preset/create

Method: POST

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
serial	integer	Yes		Preset ID number	
name	string	Yes		Scene name: Within 64 bytes (each Chinese character occupies 2 bytes)	
loadType	integer	Yes		Preset loading method: 1: Complete. 2: Relative.	
type	integer	Yes		Preset type: 1. Global preset.	

Name	Type	Required	Default Value	Remarks	More
				2. Single-screen preset. 3. Multi-screen preset. 4. Screen preset (screen template).	
sourceRegion	integer	Yes		Preset saved from PVW or PGM 2: PGM. 4: PVW.	
switchEffect	object	Yes		The global transition effect	
└─ time	number	Yes		The transition duration (unit: ms)	
└─ type	number	Yes		Transition effects: 0: Cut. 1: Fade. 2: Dip. 3: Wipe. 4: Stinger. 5: DVE.	
└─ wipe	object	No		Wipe	
└─ image	integer	No		Pattern settings: 1. Vertical rectangle: The height remains constant while the width increases from left to right, and decreases in the opposite direction when reversed. The border appears only	

Name	Type	Required	Default Value	Remarks	More
				on the moving edge (middle vertical line). 2. Horizontal rectangle: The width remains constant while the height increases from top to bottom, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle horizontal line). 3. Vertical rectangle: The height remains constant while the width increases from the center towards both sides, and decreases in the opposite direction when reversed. The border appears only on the moving edge (vertical lines moving outward from the center). 4. Horizontal rectangle: The width remains constant while the height increases from top to	

Name	Type	Required	Default Value	Remarks	More
				bottom, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle horizontal line). 5. Center rectangle. 6. Top-left rectangle. 7. Top-right rectangle. 8. Bottom-right rectangle. 9. Bottom-left rectangle. 10. Top-center rectangle. 11. Right-center rectangle. 12. Bottom-center rectangle. 13. Left-center rectangle. 14. Cross shape. 15. Center diamond. 16. Center ellipse. 17. Top-left triangle. 18. Top-right triangle.	
 symmetry	integer	No		Symmetry, ranging from 0% to 100%, with a coefficient of 100 (needs to be multiplied by 100)	

Name	Type	Required	Default Value	Remarks	More
└─ posX	integer	No		The X position where the pattern appears	
└─ posY	integer	No		The Y position where the pattern appears	
└─ frameW	integer	No		Border width	
└─ frameS	integer	No		Border softness	
└─ direction	integer	No		Direction: 1. Forward. 2. Reverse. 3. Switch.	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source.	

Name	Type	Required	Default Value	Remarks	More
				13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ dve	object	No		dve	
└─ type	integer	No		Type: Type: 1. Push. 2. Squeeze. 3. Material effect.	
└─ image	integer	No		Image: Based on the UI, from left to right and top to bottom.	
└─ brightness	integer	No		Clip	
└─ softness	integer	No		Softness	
└─ directionDev	integer	No		Dve direction: 0. Upper left 1. Up 2. Upper right 3. Right 4. Lower right 5. Down 6. Lower left 7. Left	
└─ directionEffect	string	No		Effect direction: 1. Forward. 2. Reverse. 3. Switch.	

Name	Type	Required	Default Value	Remarks	More
└─ keySource	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19: Cleanfeed.	
└─ KeySourceReverse	integer	No		Whether to enable invert colors for the key source: 0: Disable.	

Name	Type	Required	Default Value	Remarks	More
				1: Enable.	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ stinger	object	No		stinger	
└─	integer	No		Playback start time	

Name	Type	Required	Default Value	Remarks	More
playStartTime					
└─ mixtureTime	integer	No		Mix time	
└─ mixtureStartTime	integer	No		Mix start time	
└─ matting	integer	No		DSK switch: 1: Enable. 0: Disable.	
└─ brightness	integer	No		Clip	
└─ softness	integer	No		Softness	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source.	

Name	Type	Required	Default Value	Remarks	More
				14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ dip	object	No		Dip	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD.	

Name	Type	Required	Default Value	Remarks	More
				16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
createDefault	integer	No		Create default preset data (for central control device and LCD use): 0: No. 1: Yes.	
params	string []	Yes		Preset parameter function enumeration	Item type: string
└─		No		For enumeration details, see the preset configuration file.	
screens	object []	Yes		Screen data in the preset	Item type: object
└─ guid	string	Yes		Screen ID	
└─ layers	object []	Yes		Layer parameters in the preset	Item type: object
└─ serial	integer	Yes		Layer No. (required)	
└─ type	integer	Yes		Layer type (required)	
interfaces	integer []	No		Connector ID	Item type: integer

Name	Type	Required	Default Value	Remarks	More
└─		No			

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	number	No		Response code	
message	string	No		Response message	
data	object	No			
└─ guid	string	Yes		Preset ID	
└─ serial	integer	Yes		Preset ID number	
└─ name	string	Yes		Scene name: Within 64 bytes (each Chinese character occupies 2 bytes)	
└─ loadType	integer	Yes		Preset loading method: 1: Complete. 2: Relative.	
└─ type	integer	Yes		Preset type: 1. Global preset. 2. Single-screen preset. 3. Multi-screen preset.	
└─ sourceRegion	integer	Yes		Preset saved from PVW or PGM 2: PGM. 4: PVW.	
└─ switchEffect	object	Yes		The global transition effect	
└─ time	number	Yes		The transition	

Name	Type	Required	Default Value	Remarks	More
				duration (unit: ms)	
└─ type	number	Yes		Transition effects: 0: Cut. 1: Fade. 2. Dip. 3. Wipe. 4. Stinger. 5. DVE.	
└─ wipe	object	No		Wipe	
└─ image	integer	No		Pattern settings: 1. Vertical rectangle: The height remains constant while the width increases from left to right, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle vertical line). 2. Horizontal rectangle: The width remains constant while the height increases from top to bottom, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle horizontal	

Name	Type	Required	Default Value	Remarks	More
				line). 3. Vertical rectangle: The height remains constant while the width increases from the center towards both sides, and decreases in the opposite direction when reversed. The border appears only on the moving edge (vertical lines moving outward from the center). 4. Horizontal rectangle: The width remains constant while the height increases from top to bottom, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle horizontal line). 5. Center rectangle. 6. Top-left rectangle. 7. Top-right rectangle. 8. Bottom-right rectangle.	

Name	Type	Required	Default Value	Remarks	More
				9. Bottom-left rectangle. 10. Top-center rectangle. 11. Right-center rectangle. 12. Bottom-center rectangle. 13. Left-center rectangle. 14. Cross shape. 15. Center diamond. 16. Center ellipse. 17. Top-left triangle. 18. Top-right triangle.	
└─ symmetry	integer	No		Symmetry, ranging from 0% to 100%, with a coefficient of 100 (needs to be multiplied by 100)	
└─ posX	integer	No		The X position where the pattern appears	
└─ posY	integer	No		The Y position where the pattern appears	
└─ frameW	integer	No		Border width	
└─ frameS	integer	No		Border softness	
└─ direction	integer	No		Direction: 1. Forward. 2. Reverse. 3. Switch.	
└─ source	object	No			

Name	Type	Required	Default Value	Remarks	More
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ dve	object	No		dve	
└─ type	integer	No		Type: Type: 1. Push. 2. Squeeze. 3. Material effect.	

Name	Type	Required	Default Value	Remarks	More
└─ image	integer	No		Image: Based on the UI, from left to right and top to bottom.	
└─ brightness	integer	No		Clip	
└─ softness	integer	No		Softness	
└─ directionDev	integer	No		Dve direction: 0. Upper left 1. Up 2. Upper right 3. Right 4. Lower right 5. Down 6. Lower left 7. Left	
└─ directionEffect	string	No		Effect direction: 1. Forward. 2. Reverse. 3. Switch.	
└─ keySource	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image.	

Name	Type	Required	Default Value	Remarks	More
				7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
 KeySourceReverse	integer	No		Whether to enable invert colors for the key source: 0: Disable. 1: Enable.	
 source	object	No			
 general	object	No		General information	
 sourceId	integer	No		Source ID, which is unique for a specific source type	
 sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW.	

Name	Type	Required	Default Value	Remarks	More
				5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ stinger	object	No		stinger	
└─ playStartTime	integer	No		Playback start time	
└─ mixtureTime	integer	No		Mix time	
└─ mixtureStartTime	integer	No		Mix start time	
└─ matting	integer	No		DSK switch: 1: Enable. 0: Disable.	
└─ brightness	integer	No		Clip	
└─ softness	integer	No		Softness	
└─ source	object	No			
└─ general	object	No		General information	

Name	Type	Required	Default Value	Remarks	More
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ dip	object	No		Dip	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific	

Name	Type	Required	Default Value	Remarks	More
				source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ params	string []	Yes		Preset parameter function enumeration	Item type: string
└─		No		For enumeration details, see the preset configuration file.	

Name	Type	Required	Default Value	Remarks	More
└─ screens	object []	Yes		Screen data in the preset	Item type: object
└─ guid	string	Yes		Screen ID	
└─ layers	object []	Yes		Layer parameters in the preset	Item type: object
└─ serial	integer	Yes		Layer No. (required)	
└─ type	integer	Yes		Layer type (required)	
└─ interfaces	integer []	No		Connector ID	Item type: integer
└─		No			

6.2.3 Load Preset

● Basic Information

Path: /v1/preset/apply

Method: POST

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
presetId	string	No		Preset GUID	

Name	Type	Required	Default Value	Remarks	More
serial	integer	No		Preset ID	
targetRegion	integer	Yes		Area to apply: 2: PGM. 4: PVW; 0: Auto (PVW for switchers, PGM for controllers and splicers).	
auxiliary	object	Yes		Auxiliary information	
└─ keyFrame	object	Yes		KeyFrame information included when loading the preset	
└─ enable	number	Yes		Execute KeyFrame when loading the preset: 0: Yes. 1: No.	
└─ effect	object	Yes		Preset playback effect switch	
└─ enable	number	Yes		Execute effects when loading the preset: 0: Yes. 1: No.	
└─ switchEffect	object	Yes		The global transition effect	
└─ time	number	Yes		The transition duration (unit: ms)	
└─ type	number	Yes		Transition effects: 0: Cut. 1: Fade. 2: Dip.	

Name	Type	Required	Default Value	Remarks	More
				3. Wipe. 4. Stinger. 5. DVE.	
└─ wipe	object	No		Wipe	
└─ image	integer	No		Pattern settings: 1. Vertical rectangle: The height remains constant while the width increases from left to right, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle vertical line). 2. Horizontal rectangle: The width remains constant while the height increases from top to bottom, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle horizontal line). 3. Vertical rectangle: The height remains constant while the width increases from	

Name	Type	Required	Default Value	Remarks	More
				the center towards both sides, and decreases in the opposite direction when reversed. The border appears only on the moving edge (vertical lines moving outward from the center). 4. Horizontal rectangle: The width remains constant while the height increases from top to bottom, and decreases in the opposite direction when reversed. The border appears only on the moving edge (middle horizontal line). 5. Center rectangle. 6. Top-left rectangle. 7. Top-right rectangle. 8. Bottom-right rectangle. 9. Bottom-left rectangle. 10. Top-center rectangle. 11. Right-center rectangle.	

Name	Type	Required	Default Value	Remarks	More
				12. Bottom-center rectangle. 13. Left-center rectangle. 14. Cross shape. 15. Center diamond. 16. Center ellipse. 17. Top-left triangle. 18. Top-right triangle.	
└─ symmetry	integer	No		Symmetry, ranging from 0% to 100%, with a coefficient of 100 (needs to be multiplied by 100)	
└─ posX	integer	No		The X position where the pattern appears	
└─ posY	integer	No		The Y position where the pattern appears	
└─ frameW	integer	No		Border width	
└─ frameS	integer	No		Border softness	
└─ direction	integer	No		Direction: 1. Forward. 2. Reverse. 3. Switch.	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer.	

Name	Type	Required	Default Value	Remarks	More
				1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ dve	object	No		dve	
└─ type	integer	No		Type: Type: 1. Push. 2. Squeeze. 3. Material effect.	
└─ image	integer	No		Image: Based on the UI, from left to right and top to bottom.	
└─ brightness	integer	No		Clip	
└─ softness	integer	No		Softness	

Name	Type	Required	Default Value	Remarks	More
└─ directionDev	integer	No		Dve direction: 0. Upper left 1. Up 2. Upper right 3. Right 4. Lower right 5. Down 6. Lower left 7. Left	
└─ directionEffect	string	No		Effect direction: 1. Forward. 2. Reverse. 3. Switch.	
└─ keySource	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source.	

Name	Type	Required	Default Value	Remarks	More
				12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
 KeySourceReverse	integer	No		Whether to enable invert colors for the key source: 0: Disable. 1: Enable.	
 source	object	No			
 general	object	No		General information	
 sourceId	integer	No		Source ID, which is unique for a specific source type	
 sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic	

Name	Type	Required	Default Value	Remarks	More
				screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ stinger	object	No		stinger	
└─ playStartTime	integer	No		Playback start time	
└─ mixtureTime	integer	No		Mix time	
└─ mixtureStartTime	integer	No		Mix start time	
└─ matting	integer	No		DSK switch: 1: Enable. 0: Disable.	
└─ brightness	integer	No		Clip	
└─ softness	integer	No		Softness	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer.	

Name	Type	Required	Default Value	Remarks	More
				1: No-source layer. 2: Input type. 3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ dip	object	No		Dip	
└─ source	object	No			
└─ general	object	No		General information	
└─ sourceId	integer	No		Source ID, which is unique for a specific source type	
└─ sourceType	integer	No		Source type: 0: Empty layer. 1: No-source layer. 2: Input type.	

Name	Type	Required	Default Value	Remarks	More
				3: PGM. 4: PVW. 5: BKG image. 6: LOGO image. 7: IPC type. 8: Cropped source type. 10: IPC mosaic screen. 11: Internal source. 12: Internal shape source. 13: USB drive source. 14: Image OSD. 15: Text OSD. 16: Volume Unit Meter. 17: Streaming. 18: Recording. 19. Cleanfeed.	
└─ swapEnable	number	Yes		Swap switch: 0: Disable. 1: Enable.	
└─ screens	object []	No		List of target screens for preset loading.	Item type: object
└─ guid	string	No		Screen ID	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	number	Yes		Response code	
message	string	Yes		Response message	

Name	Type	Required	Default Value	Remarks	More
data	object	Yes			

6.2.4 Rename Preset

● Basic Information

Path: /v1/preset/update-name

Method: POST

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
	object { }	No			Item type: object
└─ guid	string	Yes		Preset ID	
└─ oldName	string	No		Old preset name	
└─ name	string	Yes		New preset name	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	number	No		Response code	
message	string	No		Response message	
data	object	No			

6.2.5 Delete Preset

● Basic Information

Path: /v1/preset/delete

Method: POST

Description:

● Request Parameters

Headers

Name	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default Value	Remarks	More
	string []	No			Item type: string
└─		No		Preset GUID list	

● Return Data

Name	Type	Required	Default Value	Remarks	More
code	integer	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			

7. Gallery

7.1 Retrieve Image Details

- **Basic Information**

Path: /v1/picture/list

Method: GET

Description:

- **Request Parameters**

Query

Name	Required	Example	Remarks
limit	No		Items per page, ignored if this parameter is absent.
page	No		Query page number, ignored if this parameter is absent.
type	No		Image type: defined in the same way as the type definition in pictureObj

- **Return Data**

Name	Type	Required	Default Value	Remarks	More
code	number	Yes		Response code	
message	string	Yes		Response message	
data	object	Yes			
└─ list	object []	Yes			Item type: object
└─ pictureId	number	Yes		Image file ID	
└─ isUsed	number	Yes		Image usage status: 1: In use. 2: Not in use.	
└─	object	Yes			

Name	Type	Required	Default Value	Remarks	More
pictureObj					
└─ id	number	Yes		Sequence number under this type	
└─ order	number	Yes		Order under this type	
└─ type	number	Yes		// 0: Temporary type. 1: BKG. 2: LOGO. 3: OSD. 4: Clock. 5: Internal source.	
└─ general	object	Yes			
└─ name	string	Yes		File name	
└─ width	number	Yes		Image width	
└─ height	number	Yes		Image height	
└─ size	number	Yes		Image size, unit: KB	
└─ path	object []	Yes			Item type: object
└─ type	number	Yes		0: Thumbnail. 1: Original image (bmp).	
└─ url	string	Yes		URL path of the image	
└─ relativePath	string	Yes		Relative path of the image (used by LCD)	
└─ position	number	Yes		BKG saved position	
└─ usedCapacity	number	Yes		Float type, capacity used by the gallery (MB)	

Name	Type	Required	Default Value	Remarks	More
└─ stateInfo	object	Yes		Gallery module status	
└─ import	number	Yes		0: Idle. 1: Importing.	
└─ totalCapacity	number	Yes		Float type, total capacity of the gallery (MB)	
└─ count	number	Yes		Number of items on this page	
└─ page	number	Yes		Current page number	
└─ totalCount	number	Yes		Total number of items	
└─ totalPage	number	Yes		Total pages	